User Manual

Quick Data Loader & Tagetik Data Loader

This manual provides an overview of Quick Data Loader and Tagetik Data Loader, describing their configuration and use for importing and managing data in CCH Tagetik.

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Table of contents

The Quick Data Loader 1

The Quick Data Loader function 1

Import data via Quick Data Loader 4

Compliance with GDPR/DAC6 6

Quick Data Loader - source/target definition window 6

Quick Data Loader - Import criteria definition window 10

Quick Data Loader - Editability check outcome window 14

Quick Data Loader - Datasource mapping page 15

Quick Data Loader window: outcome 16

Working with Quick Data Loader 17

Intelligent Data Loader - Import criteria definition window 20

The Quick Data Loader in Analytical Workspaces 24

Loading data from external datasources onto Analytical Workspace 24

Uploading data from Analytical Workspace to Financial Workspace 25

Reading and writing data on Analytical Workspace 27

Tagetik Data Loader - overview 29

Tagetik Data Loader - architecture 30

Tagetik Data Loader 8 - Installation and Configuration 31

Tagetik Data Loader 9 - Installation and Configuration 38

Installation 38

Configuration 38

Java Home configuration 38

Starting Karaf 43

Tagetik Data Loader - Registration and monitoring on CCH Tagetik 44

The Quick Data Loader

Contents

This section covers the following topics:

The Quick Data Loader function 1
Import data via Quick Data Loader 4
Compliance with GDPR/DAC6 6
Quick Data Loader - source/target definition window 6
Quick Data Loader - Import criteria definition window 10
Quick Data Loader - Editability check outcome window 14
Quick Data Loader - Datasource mapping page 15
Quick Data Loader window: outcome 16
Working with Quick Data Loader 17
Intelligent Data Loader - Import criteria definition window 20

The Quick Data Loader function

Introduction

The *Quick Data Loader* is a utility which allows you to enter data CCH Tagetik quickly, en masse. In particular, it allows you to import data from the following sources into the Financial Workspace data structures and Analytical Workspace datasets:

- one or more spreadsheets or text files
- Analytical Workspace datasets
- data tables from the CCH Tagetik database
- external sever (through APO RESTful services with OData protocol)

Data entered via the Quick Data Loader is subject to the same loading checks as for data entry (user rights, editability privileges etc.).

Access to the Quick Data Loader function

The Quick Data Loader function can be accessed via the following entry points:

- the consolidator cockpit
- Analytical Workspace datasets

Quick Data Loader data processing can be initiated through:

- application jobs
- process task workflow jobs
- the Home tab of web forms launched in data entry mode (see La pagina web form (Data entry))

Data loaded via the Quick Data Loader are marked as originating from the Quick Data Loader.

Data import steps

Data is imported via the Quick Data Loader by working in various windows/pages according to the following steps:

Steps	Window/page
Definition of the data source and the target and uploading of the source into the Quick Data Loader	"Quick Data Loader - source/target definition window" on page 6
Mapping of data from the source on the target information model and definition of the calculation and data management criteria	"Quick Data Loader - Import criteria definition window" on page 10

Steps	Window/page
Data import	-
Displaying errors on blocked data in editability checks	"Quick Data Loader - Editability check outcome window" on page 14
Displaying the import outcomes	"Quick Data Loader window: outcome" on page 16

Data mapping

To run the data import, it is necessary to run the mapping between the data from the source with the structure and data from the target. When the Quick Data Loader is accessed via the contributor cockpit, the mapping operation must be executed manually. However, when the mapping is accessed from Analytical Workspaces, during the step of loading data from the source, the system will compare the source dimensions with the contextual dimensions at the entry point and run an automatic mapping of the consistent data. For example, if the source has scenario, period and entity dimensions that are identical to those included in the dataset from which the mapping window was opened, the system automatically runs the mapping of the three identical dimensions.

You can then modify the automatic mapping, by changing or removing the related dimensions, and manually add to the target one or more dimensions that are only present in the source.

If the source is an Analytical Workspace type source, the data is also mapped based on whether the target of the dataset's analytical dimensions are included in the list and in the user rights.

Dynamic mapping

Dynamic mapping criteria can be created when data is loaded into Analytical Workspace and Financial Workspace. At the start of the data processing, these criteria calculate the data to be entered for each dimension, based on the execution context.

The possibilities for creating dynamic mapping criteria are as follows:

Element	Description
Default function	Defines the dimension data during the data extraction step. The function can be defined using the following criteria: • a constant • a context parameter (*#PROCESS#*, *#SCENARIOS#*, *#PERIODS#*, *#ENTITIES#*, *#REFERENCE_DATE#*, *#TARGET_PARTITION#*, *#USER#*, *#AW_CODE#*) • an Analytical Workspace parameter (which can be called using the syntax *@parameter name@*) • a parameter related to dimensional attributes (which can be called using the syntax *${ parameter }*) • a unified SQL syntax (see Sintassi SQL unificata) • a {FILE_NAME} parameter relating to the name of the source file ***Note**: In the case of a single file source then the displayed file name is the concatenated 12-character random NAMEFILE.* • a {FILE_PATH} parameter relating to the path to the source file **Note**: in the case of a single file source, the value will not be defined.
SQL function	Defines the data dynamically during the extraction and can apply small transformations. The definition of the function is alternative to the definition of a default value; the latter is left for backward compatibility reasons.

When dynamic mapping is created on a source/target relationship, the saved criteria are automatically applied in all data processing on the specific relationship.

Conversion of periodic amounts into progressive amounts

When the import destination is a gross or intercompany amount, but the amounts in the source are expressed in a periodic format (on a monthly, bi-monthly, etc. basis), the Quick Data Loader converts the amounts to a progressive format as required by the Financial Workspaces of CCH Tagetik . This calculation is applied to P&L accounts and other flows, or to variation accounts.

In single-submission processes, data loaded from the data source are made progressive without being enriched by data from CCH Tagetik . Depending on the period length, any periods needed, and not present

on the target, are generated. The calculation is based on the period length indicated in the related dimension. For example, if the period length is bimonthly, progressive values are generated for periods 02, 04, 06...; or if the period length is half-yearly, progressive values are generated for periods 06, 12.

In the scenario/period submission processes, the periodic data of the loaded scenario/period are enriched with the progressive scenario/period data already present in CCH Tagetik . This calculation considers the period length and the carry forward policies indicated in the Quick Data Loader's data management criteria.

Data for the previous scenario/period read by CCH Tagetik will depend on the type of data imported (Gross/intercompany) and also on the type of amounts present in the source (gross/minority).

Particular features of initiating via web forms

When the Quick Data Loader data process is initiated from a data entry web form, the target type depends on the matrices present in the web form. With matrices created in Financial Workspace on one or more sheets, the only type of target available is Data (Amounts, Intercompany). With matrices created in Analytical Workspaces on one or more sheets, the only type of target available is Analytical Workspace. When the form contains matrices in both Financial Workspace and Analytical Workspace, it is possible to choose the desired type of target.

Particular features of dimensions

Below are some dimensions that are only visible if the **Target Type** has been set to *Data (Balances, Intercompany)*):

Element	Description
Period Length	This allows you to indicate whether the period over which the source amounts are calculated is cumulative (adding amounts), or periodic (monthly, bimonthly, etc.).
Entity currency	Indicates the entity currency This dimension is not visible because it is calculated automatically based on the entity currency for the process being run.
Entity currency amount	Indicates the value of the amount of the transaction converted into the entity currency. If not mapped to a source column, it is automatically calculated as a conversion of the transaction amount.
Amount	Indicates an amount type data. For this dimension, it is not possible to specify a default value. Relating several columns from the source to this dimension will activate the *Measures* mode, which allows you to: • define a list of dimensions from the template, typically not present in the source, on which the related measures will be based • indicate, for each measure, the values that the template dimensions must assume

Default Value

For some dimensions the defaults are already set:

Element	Default
Scenarios	• in the case of a submission process for period scenario, the context scenario from which the Quick Data Loader is opened
Period	• in the case of a submission process for period scenario, the context period from which the Quick Data Loader is opened
Entity	the entity of the context from which the Quick Data Loader is opened (in the case of opening from the process cockpit, only in the case of a single entity selected)
Category	the amount system category $AMOUNT
Custom dimensions	the default targets of the entity in case of opening from a single entity otherwise the default custom dimensions defined in configuration
Measures (Dataset fields of type Text and Number)	• default set in the definition of the relevant Dataset field.

The default cannot be set manually, but only by selection from the relevant value range.

Import data via Quick Data Loader

Introduction

By importing data via Quick Data Loader, the following activities are performed simultaneously:

- data transfer from source to target
- Data editability check

Steps of data import into target (Amounts, Intercompany)

The data import into target (Amount, Intercompany) takes the following steps:

1. The records in the selected data source are read.
2. If Measures mode is enabled, the Amount dimension is defined (by pivoting) with the values on the columns mapped as Measures, while the dimensions on which the measures move are defined with the default values.
3. Values are calculated for dimensions that are not directly populated in the data source.
4. Records that do not satisfy the filters inherited from the execution context (see "Data mapping" on page 2) are deleted.
5. If the period length indicates exact amounts, conversion to progressive amounts is performed (see Transformation of"Conversion of periodic amounts into progressive amounts" on page 2unts).
6. An editability check of the data to be deleted and/or inserted is performed.
7. The records in CCH Tagetik suitable for control are modified.

Editability checks on targets (Amounts, Intercompany)

If, during the data editability check, the system detects anomalies in the data types to be loaded and/or in the loading process, the processing is interrupted and the detail of the records causing the warnings is displayed (see "Quick Data Loader - Editability check outcome window" on page 14).

Importing data to an Analytical Workspace target

Data import to an Analytical Workspace target follows the following steps:

1. Reading of all record in the selected data source.
2. Deleting records that do not meet the filters inherited from the execution context (see "Data mapping" on page 2).
3. Editability check of data to be deleted and to be inserted (depending on the loading mode)
4. Input on the dataset of records satisfying these checks with application of the defaults set on the measure type fields if not managed as input.

Editability checks on the Analytical Workspace target

Below is the list of editability checks for the Analytical Workspace targets:

- dataset with at least one measure belonging to a step greater than or equal to the execution step
- existing, active and visible scenario for the current user
- existing scenario/period, not blocked, and belonging to the run process
- existing Entity, belonging to the run process and editable for the current user
- Existing Entity node
- existing account, belonging to the process and visible to the current user
- Existing Account node
- existing category, belonging to the process and visible to the current user
- account (or account/category) belonging to the process and to a step greater than or equal to the run step
- existing custom dimensions visible to the current user
- Existing custom dimension node
- existing and editable Ctp Entity for the current user
- existing and editable Ctp Custom dimension for the current user
- existing currency
- existing variation/detail type
- existing and visible analytical dimensions for the current user
- Existing analytical dimension node
- existence analysis of duplicate keys, both in the source data only and in the simulation of the situation after the deletion and insertion steps

Standard Process

- Entity belonging to the run process and not submitted for the current phase
- Entity node not submitted for the current step

Multidimensional Process

- Dimension element of the current step belonging to the run process and not submitted for the current step. Elements of the dimensions linked to the other unsubmitted steps in the step to which they belong.
- Unsubmitted node of the dimension of the current step. Nodes of the dimensions linked to other unsubmitted steps in the step to which they belong.

If during the data editability check the system detects anomalies, the data processing is interrupted and the detail of the records causing the warnings is displayed (see "Quick Data Loader - Editability check outcome window" on page 14).

Restriction checks on Analytical Workspace targets

Dimensions-related visibility checks for the current user are only performed if the relevant dimension is present on the target dataset.

Process checks for Analytical Workspace targets

Process and step checks are only carried out when, in addition to the presence of account, category and dataset information, it is possible to uniquely derive scenario, period and Entity information compatible with the partitioning of the target dataset:

Partitioning type	Required conditions	
Unpartitioned Datasets	Scenario, Period and Entity columns are required	
Datasets partitioned by business cycle	**If the execution was carried out by...**	**Then the following is necessary...**
	a submission process for Scenario/Period	the Entity column (Scenario and Period columns are optional)
	a single submission process	Scenario, Period and Entity columns
Datasets partitioned by business cycle and entity	**If the execution was carried out by...**	**Then the following is necessary...**
	a submission process for Scenario/Period by selecting a single Entity	none of the columns Scenario, Period and Entity, which are optional
	a submission process for Scenario/Period by selecting an Entity Node or more than one Entity,	the Entity column
	a single submission process by selecting a single Entity	the Scenario and Period columns
	a single submission process by selecting an Entity node or more than one Entity	Scenario, Period and Entity columns

Compliance with GDPR/DAC6

Data processing criteria compliant with GDPR/DAC6

In applications compliant with GDPR/DAC6 legislation, the Quick Data Loader processes the data as follows.

Data source	Target type	Data processing type
External (uploaded file, Odata etc.)	Data (Amounts, Intercompany)	Data from external sources (Files, Other databases, OData services) are all assumed to be plain data (not encoded or encrypted). No data is encoded or encrypted since no current output dimension provides for encoding or the protection of personal data.
Internal query		The selection of plain data is delegated to the parametrizer.
Analytical Workspaces		A check will block the execution of the input fields with the *Encode* or *Personal data protection* options enabled.
External source (uploaded file, OData etc.)	Analytical Workspaces	the data will be encoded and/or encrypted if any output fields have the *Encode* and/or *Personal data protection* option enabled.
Internal query		The selection of data from the source, compatible with the *Encode* and *Personal data protection* options defined in the target, is delegated to the parametrizer.
Analytical Workspaces		A check will block execution if a piece of data from the source and one from the target, which have been related, have different values for the *Encode* and *Personal data protection* options.

***Note**: the preview window for the data to be imported displays the plain data even when it is coded or encrypted.*

Quick Data Loader - source/target definition window

User| <application database>

Processes tile > select process > Contributor > select Details overview/step tab > Quick Data Loader

Processes > select process > Contributor > select Details overview/step tab > Quick Data Loader

Purpose of window

This window allows you to set up in the Quick Data Loader the source from which data come and the target into which they are to be inserted.

Fields in window

Field	Description
Source type	This allows you to choose the data source to be entered in the target. • **Analytical Workspace**: dataset of an Analytical Workspace ◦ **File**: spreadsheet or text file ◦ **Databases**: databases configured on CCH Tagetik or on a Tagetik Data Loader ◦ **OData web service**: RESTful API service with OData protocol ◦ **REST API**: RESTful API service ◦ SharePoint**:** file retrieved from SharePoint site ◦ **Google BigQuery**: data on Google BigQuery platform ◦ **Snowflake**: Snowflake database ◦ MS Fabric: Wareouse
Target type	Allows you to select the **Target type**, namely the target into which data are to be inserted. • **Data (Amounts, Intercompany)**: loads data onto the value tables of the Financial Workspaces of CCH Tagetik • **Analytical Workspace**: loads data into an Analytical Workspace dataset.

Options for the Analytical Workspace source

Element	Description
Analytical Workspaces	Select an Analytical Workspace.
Dataset	Select a dataset from the selected Analytical Workspace. ***Note**: the system only displays datasets for which the user is enabled.*
Typology	Select a logical set of data of the selected dataset.

Options for the File Source

Element	Description
File type	Selects the source file format. Depending on the format, only certain file extensions will be accepted. • **Excel (.xls)** Accepted extensions [".xls", ".zip"] • **Excel (.xlsx)** Accepted extensions [".xlsx", ".xlsm", ".zip"] • **Text with separator** Accepted extensions [".txt", ".csv", ".tsv", ".zip"] • **CSV** Accepted extensions [".csv", ".zip"]
Columns separator	Only for **Text with separator** type. Select the column separator character of the source file. • ,: (default value), • *;* • *|* • *Tab*
Header	Only with **File Type** on **Text with separator**. Indicates the presence of a header row in the source file. In is unselected by default.
Value delimiter	Only with **File Type** on **Text with separator**. Select the character delimiting the values in the source file. • **None** (by default) • " • '

Element	Description
Value delimiter mode	Only for **Text with separator** type. Select the mode in which values are delimited by the delimiter character. • **None**: no value is delimited (by default) • **Minimal**: only values with separator character are delimited • **Non numeric**: only non numeric values are delimited • **All**: all values are delimited
Charset	Only for **Text with separator** type. Select the characters encoding type by which the file was created. The default value is **(Automatically detected)**.
File location	By **File** type only. Select the import mode of the source file. • **Upload**: imports from file system • **Current server**: imports from a directory visible from the application server on which CCH Tagetik is installed. • an **Endpoint** among those defined in the application of **URL** or **Tagetik Data Loader** type with source type **File**: allows a file to be loaded from a remote resource defined via an **Endpoint**.
Recursive extraction	The option is only visible with **File Origin** set to **Upload**. If selected, loading a .zip file will process or consider for preview all files located in the root folder, subfolders and all nested .zip files.
Select file to import	Only with **File Location** on **Upload**. Opens the file system to select the file and loads it automatically. ***Note**: when loading very heavy files, the progress is displayed in a pop-up .* ***Note**: It is also possible to select a file with a .zip extension in order to upload several files at the same time.* *Each time the values of the fields **Source Type**, **File Type** and File Location are modified, the previously uploaded file will be deleted.*
File path	Only with **File Location** on **Upload**. Defines the path to the desired file. • Empty: the system selects the directory defined as Writable folder in the application server's system properties. • absolute path to the directory> which must be listed among those permitted (see Pagina impostazioni Sicurezza). • <relative path from the Writable folder>
Directory	Only with **File Location** on an **Endpoint**. optional, relative path from the directory configured on the Endpoint;
File name	only with **File Source** on **current Server** or on an **Endpoint**. Specifies the specific file from which the data is read. • Name of one or more source files, together with extension, separated by the character ";". • ***: all files of the type defined in **File type** present in the directory. • A filter syntax: e.g. [filename*.txt] to indicate all files whose name begins with 'filename' and ends with '.txt'. • File names and filter syntax separated by ';'. • Name of a ZIP file complete with extension.
File management type	Only with **File Location** on **Current Server** on an **Endpoint**. Select the action to be performed on the files after importing data. • **No** • **Delete** • **Store**: files are moved to a subdirectory with the name "store"

Options for the Database Source

Element	Description
Database objects	Select a query object among those defined in CCH Tagetik
Database connection	Select the database on which to execute the query selected in **Database Objects** from those defined as primary or secondary datasources in the application. The current database is selected by default.
Database endpoint	Select an endpoint of Tagetik Data Loader type (with **Database** source type) and set the execution of the selected query in **Database Objects** via the Tagetik Data Loader (see "Tagetik Data Loader - overview" on page 29). Empty by default.

OData Service Source Options

Element	Description
OData service Endpoint	Select an endpoint of OData Service type defined in CCH Tagetik .
OData service EntitySet name	Defines a collection of Entities (EntitySet). The EntitySets described in the Entity Data Model of the service root.
OData service query options	Defines the query criteria according to the OData standard.

REST API source options

Element	Description
Endpoint REST API	Select an endpoint of REST API type defined in CCH Tagetik .
REST API query options	Defines the query criteria.

SharePoint source options

Element	Description
File type	Selects the format of source file. • **Excel (.xls)** • **Excel (.xlsx)** • **Text with separator** • **CSV**
Columns separator	Only for **Text with separator** type. Select the column separator character of the source file. • **,**: (default value), • **;** • **|** • **Tab**
Header	Only for type: **Text with separator**. Indicates the presence of a header row in the source file. In is unselected by default.
Value delimiter	Only for type: **Text with separator**. Select the character which delimits the values in the source file. • **None** (di default) • **"** • **'**
Value delimiter mode	Only for **Text with separator** type. Select the mode in which values are delimited by the delimiter character. • **None**: no value is delimited (by default) • **Minimal**: only values with separator character are delimited • **Non numeric**: only non numeric values are delimited • **All**: all values are delimited
Charset	Only for **Text with separator** type. Select the characters encoding type by which the file was created. The default value is (**Detected automatically**).
Endpoint SharePoint	Select an endpoint of SharePoint type defined in CCH Tagetik.
Drive	Select a drive defined on SharePoint.
Directory	Optional, relative path from the root of the specified Drive
File name	Name of a file along with extension.

Note: Dynamic syntax is not handled

Options for the Google BigQuery Source

Element	Description
Database object	Select a query type object from those defined in CCH Tagetik. ***Note**: The database object must have the **Other Body** field filled.*
Google BigQuery endpoint	Select a Google Big Query endpoint and sets the execution of the selected query in Database Object.

Options for the Snowflake source

Element	Description
Database object	Select a query type object from those defined in CCH Tagetik. ***Note**: The database object must have the **Other Body** field filled.*
Snowflake Endpoint	Selects an endpoint of Snowflake type and sets the execution of the selected query in Database Object.

Analytical Workspace target options

Only when **Target Type** is set to **Analytical Workspace**.

Element	Description
Analytical Workspaces	Select an Analytical Workspace.
Dataset	Select a dataset from the chosen Analytical Workspace. ***Note**: the system only displays datasets for which the user is enabled.*
Typology	Select a spreadsheet or reporting table from the chosen dataset

Options for the MS Fabric source

Element	Description
Database object	Select a query type object from those defined in CCH Tagetik. **Note**: The database object must have the **Other Body** field filled.
Endpoint MS Fabric	Select an endpoint of type MS Fabric and set the execution of the selected query in Database Object.

Buttons on the page

Element	Description
Upload	(Only with **File** source type) Imports the source file.
Run	Loads the source data and opens the **Quick Data Loader** page.

Quick Data Loader - Import criteria definition window

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

Admin | <application database>

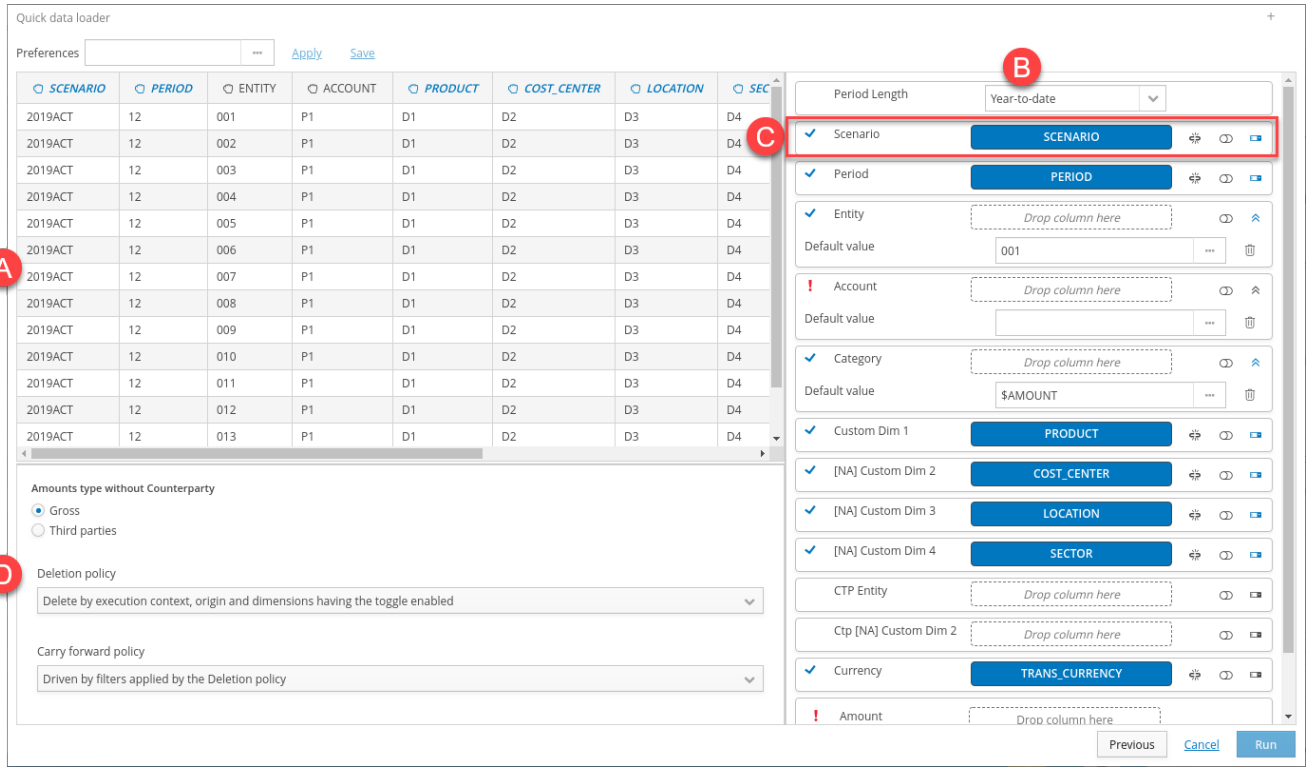
Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping or Datasources mapping

Purpose of window

On this page you can:

- Select the data from the source to be imported into the target.
- Only with target: *Data (Amounts, Intercompany)*, select the type of amount without counterparty.
- Define the criteria for overwriting or deleting data present in the database, pre-import.

What it looks like



Element	Description
To	Source data display section
B	Dimensions mapping management section
C	Dimension relationship row
D	Calculation and data management options: • Only with target: *Data (Amounts, Intercompany)*. Options for managing the type of amount calculation without counterparty. • Options for deleting data already present in databases before the data import. • Only for submission processes by scenario/period and with *Data (Amounts, Intercompany)* type target. Options for restoring data from the previous period.

Source data section

Element	Description
Data columns	Every column shows the values present on the data source, every column will have as its header: • the header that it has in the data source, if present. • the progressive numeric code of the rows in the column if the source does not generate tables with headers: • the column header, if available (only for spreadsheet or query type sources). • the dataset field header (only for Analytical Workspace type sources). ***Note**: data loaded by Analytical Workspace are filtered depending on whether the table or the user rights contain the Analytical Dimensions present in the Dataset.* ***Note**: If the data source consists of files with different headers and the header view is selected, the headers of all files will be displayed, without duplication.* A maximum of 15 rows are displayed per column.

Buttons in the source data section

Element	Description
Preferences	Allows you to enter the name of a new set of mapping preferences.
	Allows you to search for a set of preferences. The displayed preferences are those created by the current user for the current target type.
Apply	Applies the preferences of the selected set. When the following conditions are met, the preference will be applied in any case, where possible. • the name of a header saved in preferences is not present in the current data source • the default value is not present in the range of values that can be selected for the dimension in the current context A warning message informs the user of partial application.
Save	Updates or creates a set of preferences.
	Allows you to click and drag the column, or the data type, and relate it to a dimension.

Dimension mapping section

Element	Description
Dimension relationship rows	Every row shows the following: • Only with *Data (Amounts, Intercompany)* type target. Logic dimensions of CCH Tagetik Financial Workspaces. • Only with *Analytical Workspace* type target. The fields of the dataset of the Analytical Workspace selected in **Dataset**, in the Quick Data Loader's import criteria definition window (see "Quick Data Loader - source/target definition window" on page 6).

Fields for managing target dimensions

Element	Description
Period Length	This allows you to indicate whether the period over which the source amounts are calculated is cumulative (adding amounts), or periodic (monthly, bimonthly, etc.).

Buttons and icons in the dimension relationship row

Element	Description
Obligatoriness indicator	•  : mandatory dimension already managed, management can be done via relating the columns, with default values (see **Default value**) or both • ! : mandatory dimension not managed. The import cannot continue if this is not managed. • empty: optional dimension. It can be managed both by relationship and by default.
	This allows you to relate a source column to a target dimension.
 **(Toggle)**	If active, it includes the dimension as a filter in the deletion step (see "Data calculation and management options section" on the next page). ***Note**: with Data (Amounts, Intercompany)-type target, dimensions that inherit filters from the execution context have the button active by default.*
	Shows advanced options to define the dimension's default value

Element	Description
Default Value	Allows you to select a default value for the dimension. The default cannot be set manually, but only by selection from the relevant value range. ***Note**: when a Default Value is entered for a data-type dimension, it is necessary to respect the format specified in **Date Format**.* ***Note**: if the default value is defined and related to the column, during the import the system will enter any values present in the source, and replace any `null` source values with the default value.*
	Remove the dimension's default value. This is only present if the default field cannot be edited manually.
	Remove the relationship with a source column.
Date	Only with **File** or **OData** source types and data dimension types, it allows you to enter the date format with which to interpret the text originating from the external source. The entered format must match the data format of the source file and respect the java Oracle SimpleDateFormat class standard.
	Only for the **Amount** dimension in "Measures" mode. Remove the relationship with a source column.
	Only for the **Amount** dimension. Allows you to specify the values for the managed dimensions related to the measures.
	Only for the **Amount** dimension. Removes the card related to the Dimensions on which the related Measures must be moved.
 *Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping*  *Analytical Workspaces > select partition > select dataset > ⋮ > Datasources mapping*	
	Allows you to define a default function or SQL syntax to define and dynamically transform the data during data processing. See "Dynamic mapping" on page 2. ***Note**: the function definition is alternative to the **Default Value** definition.*

Data calculation and management options section

Element	Description
Amounts type without Counterparty	Only with target: **Data (Amounts, Intercompany)**. Allows you to define the data management policy when the amount type has no counterparty. • **Gross**: the system considers all amounts without a counterparty as gross amounts and amounts with a counterparty as intercompany • **3rd party**: the system considers both amounts without a counterparty and amounts with a counterparty as intercompany ***Note**: amounts with a counterparty are always also loaded as intercompany in any case.*
Deletion policy	This allows you to define the deletion policy for the data existing before the import with Quick Data Loader. • **Delete by execution context, origin and for the dimensions with toggle enabled**: for data originating from Quick Data Loader, the old data value will be deleted when new data with the same identification criteria are entered (e.g. in a context with scenario, period, entity and amount, when a new amount datum is entered on the same scenario, period, entity combination, the old amount is replaced). It also deletes dimensions with  active in the dimension relationship row. • **No deletion** • **Delete by execution context and dimensions with toggle enabled**: deletes the old data when new data with the same identification criteria is entered (e.g. in a context with scenario, period, entity and amount, when a new amount datum is entered on the same scenario, period, entity combination, the old amount is replaced). It also deletes dimensions with  active in the dimension relationship row.

Element	Description
Carry forward policy	Only for submission processes by scenario/period and with *Data (Amounts, Intercompany)* type target. Allows you to define how to reopen data from the previous period when the **period length** dimension is set to a periodic value and the imported data is not already progressive. • **Driven by imported data**: the data from the previous period will be carried forward, based on the value that the imported data has over the previous period, for the same dimensions, and will be made cumulative (without considering the origin) • **Driven by filters applied by the Deletion policy**: the data from the previous period will be carried forward, based on the value of the dimensions used as filters in the **Deletion policy** and will be made cumulative.

Buttons on the page

Element	Description
Reverse	Closes the current window without saving the data.
Previous	Where present, returns to the source and target selection step. See "Quick Data Loader - source/target definition window" on page 6.
Run	Where present, only with mandatory dimensions already added, it starts the data import.
Delete	Deletes any previously-saved mapping.
Save	Where present, it saves the mapping.

Quick Data Loader - Editability check outcome window

User | <application database>

Processes tile > select process > Contributor > select Details overview/step tab > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run > (Quick Data Loader - import criteria definition) Run

Purpose of window

This window is only displayed in the presence of warnings related to the data to be loaded.Shows the data from the source blocked by the editability checks during the Quick Data Loader's data import process.

Section on details of records excluded from being loaded into editability checks

Element	Description
With target: Data (Amounts, Intercompany)	
Loading - Gross amounts	Shows the following: • a warning on the error leading to the block on the data • the type of data and/or the type of load that led to the error • the records on which the error was detected and a message describing it
Loading - Intercompany	Shows the following: • a warning on the error that led to the block on the data • the type of data and/or the type of load that led to the error • the records on which the error was detected and a message describing it • counterparty entity • Target entity 2 counterparty
With target: Analytical Workspace	
Action	•  error detected in the data processing cancellation step •  error detected in the data processing entry step
Check	Shows the description of the check and the number of data with errors and allows you to view the detailed list by clicking on the record.

Buttons on the page

Element	Description
Previous	Returns to the import criteria definition step (see "Quick Data Loader - Import criteria definition window" on page 10).
Cancel	Closes the Quick Data Loader without saving data.
Run	Runs the Quick Data Loader's data import process, for data not blocked by editability checks.

Quick Data Loader - Datasource mapping page

Admin

Application database > Analytical Workspaces tile > workspace tile > (Dataset tab) dataset tile > ⋮ > Datasource Mapping

Application database > Navbar > Analytical Information Hub > workspace tile > (Dataset tab) dataset tile > ⋮ > Datasource Mapping

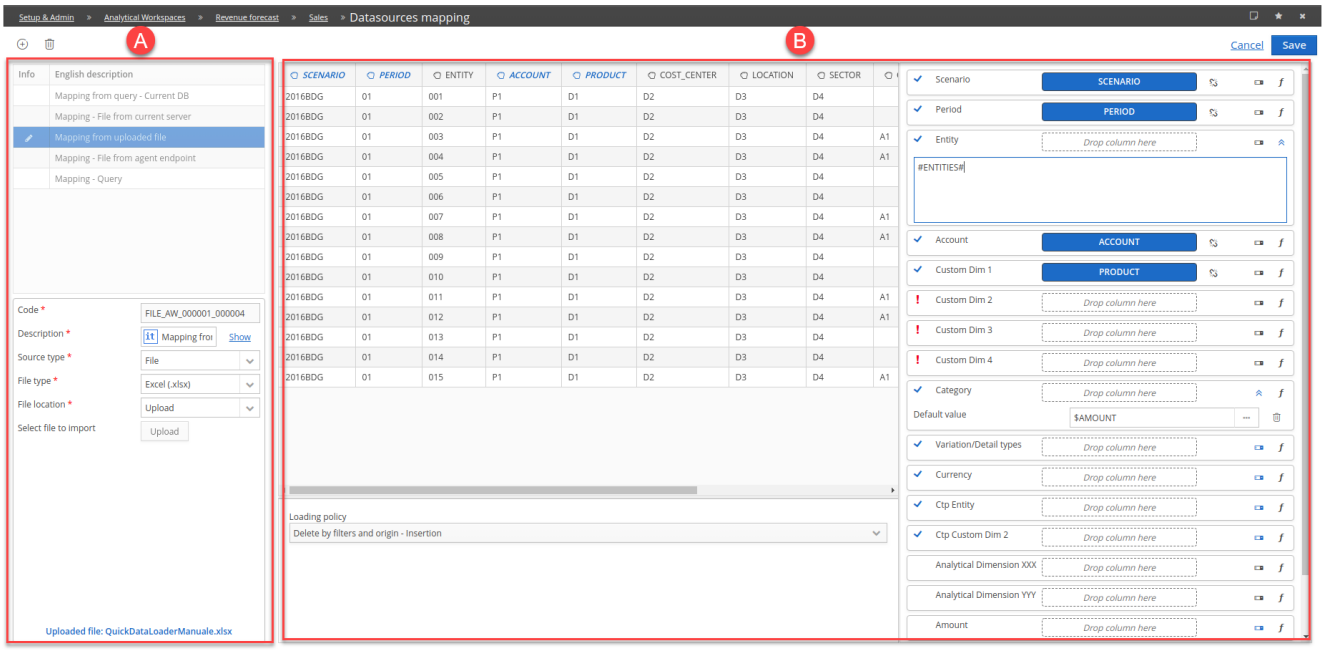
Purpose of the page

This page allows you to configure and save the mapping between the dimensions of an external source, called a *datasource*, and the fields of the Analytical Workspace dataset from which the page was opened.

This mapping is reused automatically by the Quick Data Loader when, during the data loading step, it detects the specific relationship between the datasource and the dataset of which the mapping preferences set was saved.It is typically used when the Quick Data Loader is used from the contributor cockpit, or when running Data Transformation Packages.

It can also be launched from the datasource mapping button on the Home tab of the data entry interface of the Reporting Data Entry web forms.

What it looks like



Element	Description
A	Datasource management section
B	Section for mapping datasource dimensions to the current dataset dimensions.

Buttons and icons on the page

Element	Description
	Allows you to add a datasource.
	Allows you to remove a datasource.
Automatic Mapping	Automatically maps the datasource dimensions to the current dataset dimensions.
Reset mapping	Eliminates the relationships already made for the selected datasource.
Cancel	Cancels the changes made on the selected datasource.
Save	Saves the changes made on the selected datasource.
Datasource rows	
	Allows you to change the mapping of the datasource dimensions
	Datasource added.
	Datasource removed.
Datasource management	
Upload	(Only with **File** source type) Uploads data from the source file into the mapping management section.
Load data	Uploads data from the source file into the mapping management section.

Icons in the datasource rows

Element	Description
	Datasource added.
	Datasource removed.

Fields in Datasource Management

Only the fields specific to this page are described in the following table.For more details on the source criteria definition fields, see "Quick Data Loader - source/target definition window" on page 6.

Element	Description
Code	Code of the mapping preferences set.
Description	Description of the mapping preferences set.

Mapping management section

See "Quick Data Loader - Import criteria definition window" on page 10.

Quick Data Loader window: outcome

User | <application database>

Processes tile > select process > Contributor > select Details overview/step tab > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run > (Quick Data Loader - import criteria definition) Run > (Quick Data Loader - Editability check outcome window) Run

Application database > Navbar > Processes > select process > Contributor > select Details overview/step tab > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run > (Quick Data Loader - import criteria definition) Run > (Quick Data Loader - Editability check outcome window) Run

Page sections

Element	Description
Tasks performed	Shows the details of the tasks performed.
With target: Data (Amounts, Intercompany)	
Detail of records exlcuded from loading due to editability check failure	Shows the details of data not imported because it was blocked by the editability check.
With target: Analytical Workspace	
Loading - Editability check detail: invalid elements	Shows the details of the data not imported.

Details of tasks performed section

Element	Description
Action	Data processing steps reported in the logs: • **Extraction** • **Filters**: the filters inherited from the launch context. • **Filters - detail**: details of the filters applied. • **Transformation** • **Loading**: number of rows loaded. • **Loading - Editability check details**: details of the outcomes of the editability checks performed on the rows.
Description	Description of the data processing task
Information	Number of records processed by the task
Outcome	Task outcome: • ○: task completed successfully • △: task completed with warnings • ✖: task completed with errors

Actions in Details of tasks performed

Element	Description
With target: Data (Amounts, Intercompany)	
Selection of a **Loading** record with outcome △	Displays the list of data processing records that generated warnings and the relative messages in **Detail of records excluded from loading due to editability check failure**.
With target: Analytical Workspace	
Selection of a **Loading - Editability check detail** record with outcome △	Displays the list of data processing records that generated warnings and the relative messages in **Loading - Editability check detail: invalid elements**.

Working with Quick Data Loader

Set the data source and target

User| <application database>

Processes tile > select process > Contributor > select Details overview/step tab > Quick Data Loader

Processes > select process > Contributor > select Details overview/step tab > Quick Data Loader

1. Select the desired source type: specific configuration fields appear.
2. To load the source data into the Quick Data Loader, click on the load button (**Upload** or **Run**).
3. Select the desired target type.
4. Click on **Run**.

Configure the dimension mapping

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

Admin | <application database>

Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping or Datasources mapping

1. Relate the source dimensions to the target dimensions by dragging the columns from the data section of the source to the corresponding row of the target dimension.
2. Only with target: Data (Amounts, Intercompany), set the management of amounts without a counterparty.
3. Select the desired previous data deletion policy.
4. Only with submission processes by scenario/period and Data-type target (Amounts, Intercompany), when the **Period length** dimension is set to a non-cumulative value, define the desired previous data carry forward policy.

Define a default value for dimensions

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

Admin | <application database>

Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping or Datasources mapping

During the Quick Data Loader data processing, if a default value is indicated for the dimension, the system defines any data with a null dimension value with the indicated value.

To set a default value for a dimension, click on ▭ in the dimension row, then enter the desired value.

Set dynamic data processing criteria

Admin | <application database>

Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping or Datasources mapping

To automatically define, during the data processing, data that is not otherwise specified in the dimensions (not mapped, or which contains a null value in the datasource, and on which no default value has been entered), click on f and enter the desired logic function.

Save a set of preferences related to the mapping

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

1. Perform the dimension mapping.
2. Only with target: Data (Amounts, Intercompany), set the management of amounts without a counterparty.
3. Select the desired previous data deletion policy.
4. In **Preferences**, enter a name for the set of mapping preferences, then click on **Save**.

Apply a set of preferences related to the mapping

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

1. Click on ... next to **Preferences**.
2. Search for the name of the desired dataset in the window, then click on **OK**.
3. To apply the mapping of the selected set of preferences, click on **Apply** next to **Preferences.**

Force the deletion of data not included in the deletion policy.

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

Admin | <application database>

Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping or Datasources mapping

To include a certain dimension as a filter in the deletion when it is not included in the definition, click on the button in the dimension row.

Load data from Analytical Workspace to Financial Workspace

Admin | <application database>

Analytical Workspaces > select partitions > select dataset > ⋮ > Financial workspace mapping

***Note**: on this page, the mapping target corresponds to the Data (Amounts, Intercompany).*

1. Perform the dimension mapping.
2. Set up the management of amounts without a counterparty.
3. Set the deletion policy for the previous data.
4. Only set the carry forward policy for submission processes by scenario/period.
5. To save the synchronisation configuration, click on **Save**.
6. Set the data processing execution via the desired tool (application job, process task workflow activity, data processing when closing a data entry web form; for more details see Concetti di Business Workflow and Elaborazioni di reporting).

Load data from datasource to Analytical Workspace

Admin | <application database>

Analytical Workspaces > select partition > select dataset > ⋮ > Datasources mapping

***Note**: on this page, the mapping target corresponds to the current dataset when the window is opened.*

1. Click on ⊕: a row dedicated to mapping a new datasource will appear in the datasource management table.
2. Enter a code and a description for the new datasource mapping.
3. Select the source type and specific fields that appear when making the selection.
4. Perform the dimension mapping.
5. (Optional) Define a default value for the dimensions.
6. Define the deletion policy for the previous data.
7. To save the mapping, click on **Save**.
8. Set the data processing execution via the desired tool (application job, process task workflow activity, data processing when closing a data entry web form; for more details see Concetti di Business Workflow and Elaborazioni di reporting).

Run the Quick Data Loader data loading process

User| <application database>

Processes tile > select process > Contributor > select Overview of details/step sheet > Quick Data Loader > (Quick Data Loader - Source/target definition window) Run

1. Perform the dimension mapping.
2. Click on **Run**.

Run the data loading data processing with the Intelligent Data Loader (processes with dimension by step)

user | <application database>

From a multi-organisation process, you can run the Intelligent Data Loader (IDL).

The IDL is a redesigned version of the Quick Data Loader with a new user interface. (snippet)

Navigation Panel > Processes > select process > User Tools > Data Integration > Intelligent Data Loader

1. Select the source
2. Select the target
3. Map the dimensions
4. Set up the options
5. Click **Run**.

For further details see Intelligent Data Loader

Intelligent Data Loader - Import criteria definition window

User| <application database>

Navigation Panel > Processes > select process > User Tools > Data Integration > Intelligent Data Loader

Purpose of window

This modal window allows the Intelligent Data Loader to run in a multi-organisation process.
It is only possible to run the Intelligent Data Loader if beta functionality is enabled and if you adre in a multi-organisational process in an Entity or Custom dimension step.

This window allows you to perform the following tasks

- Select the data from the source to be imported into the target.
- Select the target.
- Define the mappings between source and target
- Define criteria for overwriting or deleting data present prior to import.

What it looks like



Element	Description
P	Source selection
P	Target selection
M	Dimensions mapping management section
R	Dimension relationship row
O	Calculation and data management options: • Only with Financial Workspace target. Options for managing the type of amount calculation without counterparty. • Options for deleting data already present in databases before the data import. • Only for processes with scenario/period submission and a Financial Workspace type target. Options for restoring data from the previous period.

Source data section

Element	Description
Data columns	Every column shows the values present on the data source, every column will have as its header: • the header that it has in the data source, if present. • the progressive numeric code of the rows in the column if the source does not generate tables with headers: • the column header, if available (only for spreadsheet or query type sources). • the dataset field header (only for Analytical Workspace type sources). ***Note**: If the data source consists of files with different headers and the header view is selected, the headers of all files will be displayed, without duplication.* A maximum of 15 rows are displayed per column.

Dimension mapping section

Element	Description
Auto Mapping	Automatically maps source columns with target columns based on name matching.
Reset	Cancel mappings performed
Dimension relationship rows	Every row shows the following: • Only with *Financial Workspace* type target. The logical dimensions of the Financial Workspaces of CCH Tagetik .

Buttons and icons in the dimension relationship row

Element	Description
Obligatoriness indicator	• ✔ : mandatory dimension already managed, management can be done via relating the columns, with default values (see **Default value**) or both • ! : mandatory dimension not managed. The import cannot continue if this is not managed. • empty: optional dimension. It can be handled either via mapping or by default.
dropdown menu	This allows you to relate a source column to a target dimension.
Use Delete Filter (Toggle)	If active, it includes the dimension as a filter when deleting (see "Data calculation and management options section" on page 13).
Show Options / Hide Options	Shows and hides advanced options to define the replacement of null values and the use of the deletion filter
Replace null values	Allows you to select the type of replacement for null values: - No: do not replace any null values - with Value: replaces null values with the default value entered in the field below <Enter Value>. - with Function: allows you to define a default function or SQL syntax to dynamically define and transform data during data processing. See "Dynamic mapping" on page 2. ***Note**: when a Default Value is entered for a data-type dimension, it is necessary to respect the format specified in **Date Format**.* ***Note**: if the default value is defined and related to the column, during the import the system will enter any values present in the source, and replace any `null` source values with the default value.*
Date	Only with **File** or **OData** source types and data dimension types, it allows you to enter the date format with which to interpret the text originating from the external source. The entered format must match the data format of the source file and respect the java Oracle SimpleDateFormat class standard.

Data calculation and management options section

Element	Description
Amounts type without Counterparty	Only with target: **Data (Amounts, Intercompany)**. Allows you to define the data management policy when the amount type has no counterparty. • **Gross**: the system considers all amounts without a counterparty as gross amounts and amounts with a counterparty as intercompany • **3rd party**: the system considers both amounts without a counterparty and amounts with a counterparty as intercompany ***Note**: amounts with a counterparty are always also loaded as intercompany in any case.*
Deletion policy	This allows you to define the deletion policy for the data existing before the import with Quick Data Loader. • **Delete by execution context, origin and for the dimensions with toggle enabled**: for data originating from Quick Data Loader, the old data value will be deleted when new data with the same identification criteria are entered (e.g. in a context with scenario, period, entity and amount, when a new amount datum is entered on the same scenario, period, entity combination, the old amount is replaced). It also deletes dimensions with active in the dimension relationship row. • **No deletion** • **Delete by execution context and dimensions with toggle enabled**: deletes the old data when new data with the same identification criteria is entered (e.g. in a context with scenario, period, entity and amount, when a new amount datum is entered on the same scenario, period, entity combination, the old amount is replaced). It also deletes dimensions with active in the dimension relationship row.
Carry forward policy	Only for submission processes by scenario/period and with *Data (Amounts, Intercompany)* type target. Allows you to define how to reopen data from the previous period when the **period length** dimension is set to a periodic value and the imported data is not already progressive. • **Driven by imported data**: the data from the previous period will be carried forward, based on the value that the imported data has over the previous period, for the same dimensions, and will be made cumulative (without considering the origin) • **Driven by filters applied by the Deletion policy**: the data from the previous period will be carried forward, based on the value of the dimensions used as filters in the **Deletion policy** and will be made cumulative.
Period Length	This allows you to indicate whether the period over which the source amounts are calculated is cumulative (adding amounts), or periodic (monthly, bimonthly, etc.).

Buttons in the modal window

Element	Description
Reverse	Closes the current window without processing the data.
Run	Runs data processing.

Table of Sources and Targets enabled in the Intelligent Data Loader

Source	Target	Status
File	Financial Workspace	Enabled on Custom dimension and Entity if beta functionality is enabled.

The Quick Data Loader in Analytical Workspaces

Contents

This section covers the following topics:

Loading data from external datasources onto Analytical Workspace 24
Uploading data from Analytical Workspace to Financial Workspace 25
Reading and writing data on Analytical Workspace 27

Loading data from external datasources onto Analytical Workspace

Introduction

This module describes the data processing that allows data from external datasources such as calculation files or databases to be loaded onto the physical partitions of the Analytical Workspace datasets in the following contexts:

- scheduling of data processing via application jobs
- integration of data processing between process workflow tasks
- execution of processing when closing a data entry web form
- inclusion of data processing in the Data Transformation Package

For the data upload to be successful, the following conditions must be fulfilled:

- the datasource mapping must exist for an Analytical Workspace dataset
- the dimensions of the dataset must have been mapped to the specific dataset and the data loading mode defined

Use in the application job

Below is the list of parameters required for configuration.

Parameter	Description
Analytical Workspace	Allows you to indicate the Workspace to which the target dataset belongs
Analytical Workspace Dataset	Allows you to indicate the target Dataset belonging to the previously selected Workspace.
Analytical Workspace Dataset typology	Allows you to specify the dataset type selected previously.
Datasource	Necessary to obtain the mapping on which both the loading mode on the Analytical Workspace and the relationship between the dimensions of the datasource and those of the dataset are defined.

All these parameters are mandatory.

The following parameters are instead necessary both to identify the partition of the dataset on which to write the data and for editability checks before uploading to the Analytical Workspace:

- **Process**, mandatory
- **Scenario**, mandatory if the selected process is scenario / period submission.
- **Period**, mandatory if the selected process is scenario / period submission.
- **Entity**, optional, cannot be defined if Entity Hierarchy and Entity Node have been defined.
- **Entity hierarchy**, optional, cannot be defined if Entity has been defined.
- **Entity Node**, optional, cannot be defined if Entity has been defined; it is mandatory if Entity Hierarchy has been defined.

***Note**: At runtime, if the datasource configuration requires the value of Aanalytical workspace or reporting parameters to be specified, a blocking error is returned that does not allow the job to be executed.*

Use in the process workflow task

Below is a list of the parameters required for setting up data processing as a task in a workflow task (see Creare un task workflow).

Parameter	Description
Analytical Workspace	Allows you to indicate the Workspace to which the target dataset belongs.
Analytical Workspace Dataset	(Mandatory) Allows you to indicate the target Dataset belonging to the previously selected Workspace.
Analytical Workspace Dataset typology	(Mandatory) Allows you to select the previously selected dataset type.
Analytical Workspace Dataset Datasource	(Mandatory) Indicates the datasource mapping belonging to the previously selected dataset on which both the loading mode and the relationship between the datasource and dataset dimensions are defined.
Process, Scenario, Period, Entity, Hierarchy and **Node**	They are automatically inherited by the execution context and are needed both for editability checks and to identify the partition of the dataset.

***Note**: At runtime, if the datasource configuration requires uploading a file, or specifying the value of Analytical Workspace or Reporting parameters, a dedicated window pops up.*

Data processing run when saving a form

Below is the list of parameters required for configuration.

Parameter	Description
Analytical Workspace Dataset typology	(Mandatory) Allows you to indicate the dataset type on which to load data.
Analytical Workspace Dataset Datasource	(Mandatory) Required to obtain the mapping on which the loading mode on the Analytical Workspace and the relationship between the dimensions of the datasource and those of the dataset are defined.
Process, Scenario, Period, Entity, Hierarchy and **Node**	They can either be inherited automatically from the form or entered manually and are needed both for editability checks and to identify the partition of the dataset.

***Note**: At runtime, if the datasource configuration requires you to specify the value of Analytical Workspace or Reporting parameters, a dedicated window pops up.*

Data processing run in a Data Transformation Package (DTP)

It is possible to run the data processing by inserting it within a Data Transformation Package either by uploading it to Analytical Workspaces (see Data Transformation Package (DTP) and Loading to Analytical Workspace) or by uploading it via Write to another dataset.

Uploading data from Analytical Workspace to Financial Workspace

Introduction

This module presents the list of parameters for loading data from an Analytical Workspace dataset onto the Financial Workspace data structure in the following contexts:

- scheduling of data processing via application jobs
- integration of data processing between process workflow tasks
- launching of data processing when closing a data entry web form

To set up synchronisation, the mapping between the information content of the Analytical Workspace and that of the Financial Workspace must be saved, so that the Quick Data Loader can automatically align the Financial workspace layer with that of the Analytical Workspace.

Use in the application job

Below is the list of parameters required for configuration.

Parameter	Description
Analytical Workspace	Mandatory by default, it allows you to specify the workspace the source dataset belongs to.
Analytical Workspace Dataset	It allows you to get the mapping on which both the loading mode and the relationship between the Dataset dimensions and the logical dimensions of CCH Tagetik are defined.
Process	It is used to identify the partition of the source dataset and for editability checks before uploading to the Financial Workspace.
Scenarios	Optional by default , it becomes mandatory if the partition of the source dataset is partitioned by business cycle and the selected process is scenario/period submission.
Period	Optional by default , it becomes mandatory if the partition of the source dataset is partitioned by business cycle and the selected process is scenario/period submission.
Entity	Optional by default.
Entity hierarchy	optional, it cannot be defined if **Entity** has been defined
Entity node	optional, it cannot be defined if **Entity** has been defined; it is mandatory if **Entity Hierarchy** has been defined.

***Note**: If the configuration requires user interaction because the value of AW or Reporting parameters must be specified, a blocking error is returned that does not allow the Job to be executed*

Use in the process workflow task

Below is a list of the parameters required for setting up data processing as a task in a workflow task (see Creare un task workflow).

Parameter	Description
Analytical Workspace	Mandatory by default, it allows you to specify the workspace the source dataset belongs to.
Dataset	(Mandatory) Allows you to indicate the dataset belonging to the previously selected Workspace from which to obtain the mapping on which both the loading mode on the Financial Workspace and the relationship between the dimensions of the dataset and those on the Financial Workspace are defined.
Process, Scenario, Period and **Entity**	They are automatically inherited by the execution context and are needed both for editability checks and to identify the partition of the dataset.

***Note**: At runtime, if the configuration requires you to specify the value of Analytical Workspace or Reporting parameters, a dedicated window pops up.*

Data processing run when saving a form

Below is the list of parameters required for configuration.

Parameter	Description
Analytical Workspace Dataset	(Mandatory) Required to obtain the mapping on which both the loading mode on the Financial Workspace and the relationship between the dimensions of the dataset and those on the Financial Workspaces are defined.
Scenario, Period and **Entity**	They are automatically inherited by the execution context and are needed both for editability checks and to identify the partition of the dataset.

***Note**: When defining the data processing to be performed, if the datasource configuration requires the value of Analytical Workspace or Reporting parameters to be specified, a dedicated panel pops up.*

Reading and writing data on Analytical Workspace

Reading conditions

The Quick Data Loader's behaviour when extracting data from Analytical Workspace depends on the following conditions:

- type of source dataset partitioning
- presence of the entity dimension column in the source dataset
- source type
- type of Quick Data Loader execution (for single node or single entity).

The general reading conditions and the relative data extraction policies are shown below.

Dataset partitioning	Reading layer	Specified dimensions	Data extraction policy
None	Reporting layer or calculation layer	Any dimension present on the layer, or none	The data is extracted from the single pre-existing partition.
By business cycle	Reporting layer		The data is extracted from the single pre-existing partition by process, scenario and period.
By business cycle	Calculation layer	Entity dimension not specified	The data is extracted from the single pre-existing partition by process, scenario and period.

The reading conditions when the extraction is initiated by an entity node are shown below.

Dataset partitioning	Reading layer	Specified dimensions	Data extraction policy
By business cycle	Calculation layer	Specified entity dimension	The data is extracted from the single pre-existing partition by process, scenario, period and selected entity node.
By business cycle and entity	Reporting layer or calculation layer	Any dimension present on the layer, or none	The data is extracted from all pre-existing partitions by process, scenario, period and entities belonging to the node.

The reading conditions when the extraction is initiated by an entity are shown below.

Dataset partitioning	Reading layer	Specified dimensions	Data extraction policy
By business cycle	Calculation layer	Specified entity dimension	The data is extracted from the single pre-existing partition by process, scenario, period and selected entity.
By business cycle and entity	Reporting layer or calculation layer	Any dimension present on the layer, or none	The data is extracted from the single pre-existing partition by process, scenario, period and entity.

Writing on Analytical Workspace

The Quick Data Loader's behaviour when inputting data on Analytical Workspace depends on the following conditions:

- type of target dataset partitioning
- presence of the column related to the entity dimension in the target dataset
- type of selected target on which to load the data
- type of Quick Data Loader execution (single node or single entity)
- the Entities that were read in the extraction step
- the Deletion Policy defined for the Quick Data Loader
- the activation of the  option on the entity dimension

The writing contexts and relative data loading policies are shown below.

Dataset partitioning	Loading layer	Specified dimensions	Data loading policy
None	Reporting layer or calculation layer	Any dimension present on the layer, or none	The data is loaded onto the single existing or created partition.
By business cycle	Reporting layer	Any dimension present on the layer, or none	The data is loaded onto the single existing or created partition by process, scenario and period.

The reading conditions when the loading is initiated by an entity node are shown below.

Dataset partitioning	Loading layer	Specified dimensions	Data loading policy
By business cycle	Calculation layer	Specified entity column	The data is loaded onto the single existing or created partition by process, scenario, period and the selected entity node.
By business cycle and entity	Reporting layer or calculation layer	Any dimension present on the layer, or none	The data is loaded separately on all existing or created partitions related to the process, scenario, period and entities belonging to the node.

The reading conditions when the loading is initiated by an entity.

Dataset partitioning	Loading layer	Specified dimensions	Data loading policy
By business cycle	Calculation layer	Specified entity dimension	The data is loaded onto the single existing or created partition by process, scenario, period and selected entity.
By business cycle and entity	Reporting or calculation layer	Any dimension present on the layer, or none	The data is loaded onto the single existing or created partition by process, scenario, period and entity.

Tagetik Data Loader - overview

Tagetik Data Loader (hereinafter "TDL") is a client that allows data to be transferred from/to CCH Tagetik to/from other systems. Data can be displayed as files of database records.

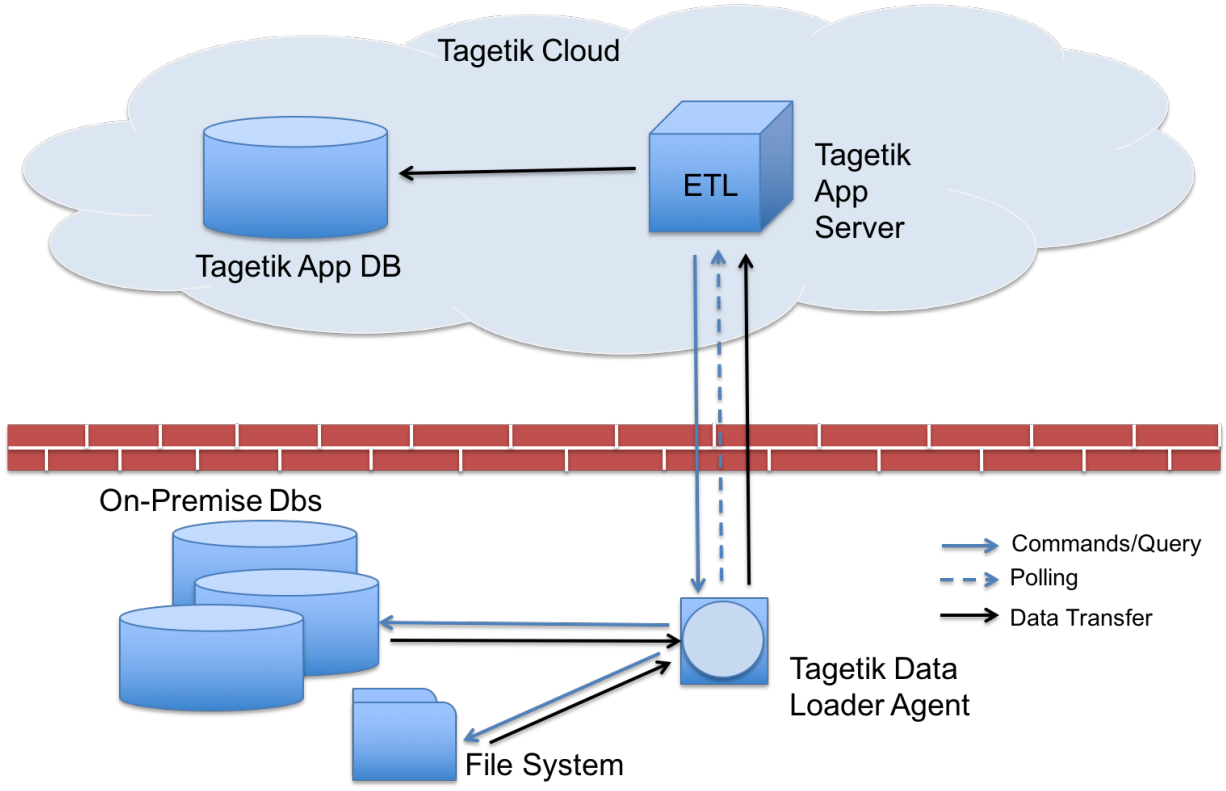
TDL is designed to be installed on-premise on the customer's network where the data is stored (private network or cloud) in order to enable data transfer via CCH Tagetik's ETL or QDL module (installed on-premise or on the cloud).

TDL allows you to select the desired transfer protocol (HTTP or HTTPS). In the case of secure data transfer (via HTTPS), TDL allows communication to take place without the customer having to create an exception in the company firewall.

Thus, Tagetik Data Loader can operate in four different types of environment (for each type of environment, TDL can transfer data in both protected mode (HTTPS) and unencrypted mode (HTTP)):

- Tagetik on-premise, TDL on-premise;
- Tagetik on-premise, TDL on cloud;
- Tagetik Cloud, TDL on-premise;
- Tagetik Cloud, TDL on cloud.

The following diagram refers to case 3.



The installation and configuration of Tagetik Data Loader involves:

- the **installation and configuration of Tagetik Data Loader**. Install the Tagetik Data Loader, configure it (connection to CCH Tagetik, logs), set up database connections, configure file transfer paths.
- **registration and monitoring of Tagetik Data Loader on CCH Tagetik**. Register, monitor and manage Tagetik Data Loader on CCH Tagetik.
- the **configuration of Endpoints on CCH Tagetik**. Enable data integration functional modules (ETL, QDL, ...) to transfer data to and from CCH Tagetik using the Tagetik Data Loader.

Tagetik Data Loader - architecture

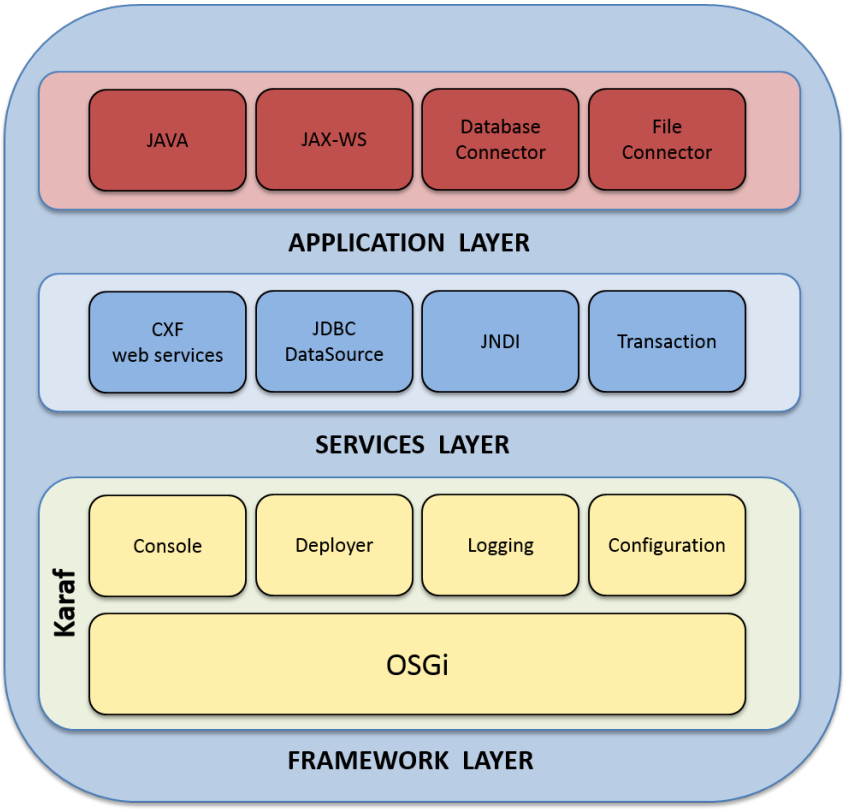
Tagetik Data Loader (hereinafter "TDL") is a programme written in Java installed and run on-premises by the customer and is able to execute tasks driven by CCH Tagetik.

TDL is structured in modules, is 'lightweight' and is dynamic, i.e. it allows modules to be installed, started, stopped and removed at runtime, thus without the need for reboots

These functionalities are provided by the use of an OSGi (Open Services Gateway initiative) framework. TDL is in fact based on Apache Karaf 4.4.0, a small OSGi framework that provides a lightweight container in which to deploy the various components and applications. Karaf 4.4.0 supports the latest OSGi R8 specification.

The TDL architecture consists of three layers as shown below:

- the first layer is the framework
- the second is the service layer
- finally, the third is the application layer



Framework layer

Apache Karaf is a small OSGi framework, which provides a 'lightweight' container in which the various components and applications can be deployed. The framework supports the following functionalities:

- **'hot update'**. Karaf supports hot deployment of OSGi bundles by tracking jar files within the [home] / deploy directory. Whenever a jar file is copied into this folder, it is installed without requiring a restart of the container
- **dynamic configuration.** The configuration of services is done using properties files within a dedicated directory. These configurations are monitored and changes to properties files are propagated to the services
- **log system.** Karaf supports a large range of logging frameworks (slf4j, log4j, etc.) that can be configured via a single file
- **administration console**. It provides an Unix-like console through which the container can be fully managed. It is also possible to extend the console and create your own customised commands.

Services layer

The service layer, located above the framework layer, interacts with the application layer to communicate with customer-developed applications that need to access and use these services. The available services are:

- **JDBC DataSource Factory**. Provides services to access databases from OSGi applications running in Karaf and to extract data from the database by installing DataSources as OSGi services
- **JNDI**. Karaf Naming (JNDI) is a feature that provides a centralised register/query function for OSGi services. JNDI API is typically used to access services from a system directory

Application layer

The Application layer is where the applications developed by Tagetik reside. It consists of java programmes that implement connectors, execute tasks and communicate with Tagetik via web services.

Tagetik Data Loader 8 - Installation and Configuration

Installation

To install the Tagetik Data Loader, simply **unzip** the file **TagetikDataLoader<*version*>_** ApacheKaraf<*version*>.***zip*** into a directory (e.g. *<custom_path>/TagetikDataLoader/*).

Version 8 of the Tagetik Data Loader uses the latest version of the Apache Karaf container, 4.4.6

Inside are two directories:

- **apache-karaf-&**lt;version>*: contains the Apache Karaf container, the Tagetik Data Loader jars and the default JDBC drivers.* The directory can be renamed to **apache-karaf**.
- **tgk-configuration-&**lt;version>: contains the externalisation of the Karaf configuration (/etc) in which the configuration files of the Tagetik Data Loader are also located.

Configuration

Tagetik Data Loader (TDL) requires the configuration of a set of parameters necessary for it to function correctly and fully:

- Java Home configuration
- definition of the other Karaf settings (memory, etc.)
- definition of Karaf configuration
- general configuration and configuration of the connection to CCH Tagetik
- authentication configuration
- logs configuration
- configuration of datasources for the databases access
- configuration of paths for accessing file system directories

Java Home configuration

In order to configure Java Home, it is necessary to access the *setenv* file(*setenv.bat* if installation is carried out on Windows operating systems) located in *<Karaf home>/bin* and define JAVA_HOME.


The JDK version to be used is 17.0.12. The Tagetik Data Loader 8 has been built and tested with JDK*17.0.12* : upgrading to subsequent patches is not discouraged in principle, as they generally contain fixes that do not disrupt compatibility. However, it should be noted that the Tagetik Data Loader is not

beforehand tested with every JDK patch beyond version 17.0.12. This occurs only in response to specific requests or reports of issues identified after a JDK update.

E.g. linux: *export JAVA_HOME="/usr/lib/jvm/openjdk-17.0.12"*
E.g. windows *set JAVA_HOME=C:\Progra~1\Java\openjdk-17.0.12*

List of tested and supported JDKs:

- Amazon Corrected 17.0.12
- Microsoft 17.0.12
- Oracle 17.0.12
- SapMachine 17.0.12
- IBM Semeru 17.0.12
- Liberica 17.0.12
- Temurin 17.0.12

Definition of other Java and Karaf settings

To define the other Java and Karaf settings, it is necessary to access the *setenv* file *(setenv.bat* if installing on Windows operating systems) located in *<Karaf home>/bin* and add the options. If Karaf is run via Windows service, after running the wrapper installation command, configure the *wrapper.conf* file available inside the configuration directory. If a custom name is specified inside the wrapper installation command, the file will be *karaf-<name>.conf*

E.g. linux:
export JAVA_MIN_MEM=256M # Minimum memory for the JVM
export JAVA_MAX_MEM=1024M # Maximum memory for the JVM
export JAVA_PERM_MEM=128M # Minimum perm memory for the JVM
export JAVA_MAX_PERM_MEM=256M # Maximum memory for the JVM

E.g. windows
set JAVA_MIN_MEM=256M # Minimum memory for the JVM
set JAVA_MAX_MEM=1024M # Maximum memory for the JVM
set JAVA_PERM_MEM=128M # Minimum perm memory for the JVM
set JAVA_MAX_PERM_MEM=256M # Maximum memory for the JVM

Karaf configuration definition

It is recommended to externalise the Karaf configuration from the *<Karaf home>/etc* folder to another directory so that all customisations can be managed when updating the Karaf container. Once outsourced, it is possible to set up the configuration directory on Karaf by accessing the *setenv* file (*setenv.bat* if installing on Windows operating systems) located in *<Karaf home>/bin* and defining the KARAF_ETC.

E.g. linux: *export KARAF_ETC="/home/user/TagetikDataLoader/tgk-configuration"*
E.g. windows *set KARAF_ETC=C:\Progra~1\TagetikDataLoader\tgk-configuration*

General setting and connection to CCH Tagetik

To configure the TDL for both the connection to CCH Tagetik and to change certain parameters relating to its functionality, it is necessary to access the *tgk.properties.cfg* file in the Karaf configuration directory.

The parameters to be configured are:

- **tgk.agent.polling.interval**. Indicates at what intervals (in seconds) the TDL verifies requests for transactions to be taken over. It is recommended not to set up a value lower than 30 seconds

- **tgk.agent.tmpfolder**. Indicates the path reserved for the TDL to save temporary resources generated by packet data transmission. The parameter is mandatory (to avoid relying on temporary system folders).
- **tgk.agent.write.db.batchsize**. Indicates the number of statements sent simultaneously to the database to be executed at write time (default 10000)
- **tgk.agent.read.db.fetchsize**. Indicates the number of records to be retrieved simultaneously from the database being read (if not specified, the default value specified by the jdbc driver in use will be used)
- **tgk.endpoint.address**. CCH Tagetik endpoint URL (e.g. https://alpha.saastagetik.com/prod)

If secure data transfer is active, i.e. the Tagetik Endpoint URL has https as protocol, in the absence of specific parameterisation, the TDL will use the default JDK trustor in use. Should you wish to specify a custom server certificate, you can do so via the following parameters:

- **tgk.server.ks.password**. Password chosen when generating the server-side certificate. It must be written following the syntax specified in Security and encryption of sensitive data
- **tgk.server.ks.path**. Complete path of the server-side certificate

If the communication between TDL and CCH Tagetik is managed by a proxy server, the following can also be configured:

- **tgk.proxy.server**. Indicates the IP of the proxy server
- **tgk.proxy.port**. Indicates the proxy server port
- **tgk.proxy.user**. Indicates the user accessing the proxy server
- **tgk.proxy.password**. Indicates the password with which the user defined in the previous property can access the proxy server. It must be written following the syntax specified in Security and encryption of sensitive data

 Starting with Tagetik Data Loader version 7.0.0, it is possible to enable payload encryption of data exchanged between the TDL and CCH Tagetik via PGP algorithm.

To do this, it is necessary to configure:

- **tgk.pgp.publickey** indicates the full path to the file containing the public key
- **tgk.pgp.privatekey** indicates the full path to the file containing the private key
- **tgk.pgp.passphrase** indicates the passphrase to use the private key. It must be written following the syntax specified in Security and encryption of sensitive data

If the Endpoint is only used to read data, only the *public key* needs to be configured; if it is only used to write data, only the secret *key* and *secret passphrase* need to be configured.
If the TDL is used to write data, only the *privatekey* with the *passphrase* on the TDL and the *publickey* on the Endpoint defined on CCH Tagetik need to be configured.

Example 1 (HTTPS secure transfer without specified custom server certificates)

```
tgk.agent.pollingInterval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=https://tagetik.installation.url.com/tagetikcpm
tgk.server.ks.path=
tgk.server.ks.password=
```

Example 1 (HTTPS secure transfer with specified custom server certificates)

```
tgk.agent.pollingInterval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
```

```
tgk.endpoint.address=https://tagetik.installation.url.com/tagetikcpm
tgk.server.ks.path=/home/TDL/certificates/server.truststore
tgk.server.ks.password={TGK_ENCRYPT}server_side_certificate_password{TGK_ENCRYPT}
```

Example 2 (HTTP transfer)

```
tgk.agent.pollingInterval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=http://tagetik.installation.url.com/tagetikcpm
```

Example 3 (PGP transfer)

```
tgk.agent.pollingInterval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=http://tagetik.installation.url.com/tagetikcpm
tgk.pgp.publickey=/home/user/TagetikDataLoade/publickey.asc
tgk.pgp.privatekey=/home/user/TagetikDataLoade/privatekey.asc
tgk.pgp.passphrase={TGK_ENCRYPT}123_pippo{TGK_ENCRYPT}
```

Authentication configuration

To enable the communication between TDL and CCH Tagetik, it is necessary to define the authentication parameters. They can be found in the file *tgk.agent.cfg* in the Karaf configuration directory.

The parameters are as follows:

- **tgk.agent.id**. Unique ID associated with the TDL instance. It is automatically generated upon the first launch of Karaf.
- **tgk.agent.secret**. SecretKey to authenticate the TDL on CCH Tagetik. It must be defined after the first launch of Karaf, once the ID has been automatically generated, and must be defined using the syntax specified in Security and encryption of sensitive data

With these two parameters it is possible to register the TDL on CCH Tagetik and authenticate it upon each request

Log configuration

The TDL log system is nothing more than a customization of the log system of Apache Karaf. The log configuration can be found in the file *org.ops4j.pax.logging.cfg* in the Karaf configuration directory and is a standard Log4j configuration file.

This file contains the following section with the specified values:

```
#Tagetik agent appender
log4j2.logger.tagetik.name = com.tagetik.cpm
log4j2.logger.tagetik.level=INFO
log4j2.logger.tagetik.additivity=false
log4j2.logger.tagetik.appenderRef.TagetikRollingFile.ref = TagetikRollingFile
log4j2.appender.tagetik.type = RollingRandomAccessFile
log4j2.appender.tagetik.name = TagetikRollingFile
log4j2.appender.tagetik.fileName = ${karaf.data}/log/tagetik.log
log4j2.appender.tagetik.filePattern = ${karaf.data}/log/tagetik.log.%i
log4j2.appender.tagetik.append = true
```

log4j2.appender.tagetik.layout.type = PatternLayout
log4j2.appender.tagetik.layout.pattern = ${log4j2.pattern}
log4j2.appender.tagetik.policies.type = Policies
log4j2.appender.tagetik.policies.size.type = SizeBasedTriggeringPolicy
log4j2.appender.tagetik.policies.size.size = 16MB

For the meaning of the various options and of the values to be assigned, refer to the Log4j specifications.

Log files are saved by default in *data/log/karaf.log*

Based on the default settings above, the TDL log file is named *tagetik.log* and is generated on a daily basis. Moreover, every day the file generated on the previous day is renamed by adding a suffix given by the production date.

Datasource configuration

In case database operations are required such as:

- reading data to be transferred to CCH Tagetik
- writing of data extracted from CCH Tagetik

TDL requires *.cfg* files to be created and configured in the Karaf configuration directory.

Here are some templates of such files:

*SqlServer (driver JTDS) - **tgk.datasource-mssql-jtds.cfg***

osgi.jdbc.driver.class=net.sourceforge.jtds.jdbc.Driver
pool=dbcp2
xa=false
url=jdbc:jtds:sqlserver://<servername>:<port>/<dbname>
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

*SqlServer (driver Microsoft) - **tgk.datasource-mssql-microsoft.cfg***

osgi.jdbc.driver.class=com.microsoft.sqlserver.jdbc.SQLServerDriver
pool=dbcp2
xa=false
url=jdbc:jtds:sqlserver://<servername>:<port>;databaseName=<dbname>
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

*Oracle - **tgk.datasource-oracle.cfg***

osgi.jdbc.driver.class=oracle.jdbc.OracleDriver
pool=dbcp2
xa=false
url=jdbc:oracle:thin:<serverName>:<port>:<serviceName>
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

PostgreSQL - ***tgk.datasource-postgresql.cfg***

osgi.jdbc.driver.class=org.postgresql.Driver
pool=dbcp2
xa=false
url=jdbc:postgresql://<serverName>:<port>/<dbName>
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

SAP Hana - ***tgk.datasource-saphana.cfg***

osgi.jdbc.driver.class=com.sap.db.jdbc.Driver
pool=dbcp2
xa=false
url=jdbc:sap://<servername>:<port>/?databaseName=<dbname>
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

IBM DB2 - ***tgk.datasource-ibmdb2.cfg***

osgi.jdbc.driver.class=com.ibm.db2.jcc.DB2Driver
pool=dbcp2
xa=false
url=jdbc:db2://<serverName>:<port>/<dbName>:retrieveMessagesFromServerOnGetMessage=true;
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

Netsuite - ***tgk.datasource-netsuite.cfg***

osgi.jdbc.driver.class=com.netsuite.jdbc.openaccess.OpenAccessDriver
pool=dbcp2
xa=false
url=jdbc:ns://<servername>:<port>;ServerDataSource=NetSuite.com;encrypted=1;CustomProperties=(AccountID=<accountId>;RoleID=<roleId>)
user=<username>
password={TGK_ENCRYPT}<userPassword>{TGK_ENCRYPT}
dataSourceName=<symbolicName>
vendor=tagetik

 The datasources passwords must be entered (both upon definition and edit) with the syntax specified in Security and encryption of sentitive data

Adding a custom datasource

From the Karaf command line via the command *service:list DataSourceFactory* it is possible to display the installed JDBC drivers.

To add a new JDBC driver it is necessary to:

- copy the **[driver_name].jar** into the *<Karaf home>/lib/endorsed* folder
- run the *install -s wrap:file:./lib/endorsed/[driver_name].jar* command from the Karaf command line
- add a *.cfg* file relating to the new datasource into the Karaf configuration directory (like those described above)
- restart Karaf

 The simultaneous presence of two JDBC drivers from the same vendor causes problems at runtime. If it is necessary to install a new JDBC driver (E.g. : a later version), the previous one has to be uninstalled.

Path configuration

In case operations are needed on files such as:

- file transfer to CCH Tagetik
- writing files produced by CCH Tagetik

TDL requires the *tgk.repository.cfg* file to be created and configured in the Karaf configuration directory.

Inside, a mapping list between the logical path name and the real absolute path can be configured.

The syntax is typical of the *.properties files, thus:

<path-1-logical-name>=<path-1-value>

<path-2-logical-name>=<path-2-value>

Security and encryption of sensitive data

For keystore and datasource passwords, there is a mechanism for encrypting the values.

To define the value of the password, the following syntax as to be used: *<password_property_name>={TGK_ENCRYPT}<password_value_to_encrypt>{TGK_ENCRYPT}*

After saving the configuration file, Tagetik Data Loader will automatically encrypt the value of the property and replace it in the same configuration file.

 TDL can only handle passwords containing 'ASCII Printable Characters', i.e. ASCII characters 32 to 126 with the following exceptions:

- The '$' character (ASCII 36) cannot be used as the last character of the password
- The character '\' (ASCII 92) cannot be used in the password.

Starting Karaf

Once installation and configuration have been carried out, Tagetik Data Loader can be used by starting Karaf via the *karaf* command(*karaf.bat* if installation is carried out on Windows operating systems) present in *<Karaf home>/bin*.

Starting as an operational system service

It is also possible to start Karaf as an OS service by executing the command *wrapper:install --name "TDL" --display "TDL" --description "TDL"* from the Karaf command line and afterwards:

- for Windows: run from the command prompt the command *TDL-service.bat install* (or *karaf-service.bat install*), present in the *<Karaf home>/bin* directory
- for Linux: follow the instructions shown once you run the *wrapper:install* command .

Tagetik Data Loader 9 - Installation and Configuration

Installation

To install the Tagetik Data Loader, simply **unzip** the file **TagetikDataLoader<*version*>_** ApacheKaraf<*version*>.***zip*** into a directory (e.g. <*custom_path*>/*TagetikDataLoader*/).

Version 9 of Tagetik Data Loader uses the latest version of Apache Karaf container, 4.4.7

Inside are two directories:

- **apache-karaf-&**lt;version>*: contains the Apache Karaf container, the Tagetik Data Loader jars and the default JDBC drivers.* The directory can be renamed to **apache-karaf**.
- **tgk-configuration-&**lt;version>: contains the externalisation of the Karaf configuration (/etc) in which the configuration files of the Tagetik Data Loader are also located.

Configuration

Tagetik Data Loader (TDL) requires the configuration of a set of parameters necessary for it to function correctly and fully:

- Java Home configuration
- definition of the other Karaf settings (memory, etc.)
- definition of Karaf configuration
- general configuration and configuration of the connection to CCH Tagetik
- authentication configuration
- logs configuration
- configuration of datasources for the databases access
- configuration of paths for accessing file system directories

Java Home configuration

In order to configure Java Home, it is necessary to access the *setenv* file*(setenv.bat* if installation is carried out on Windows operating systems) located in <*Karaf home*>/*bin* and define JAVA_HOME.


The JDK version to be used is 17.0.12. The Tagetik Data Loader 9 has been built and tested with JDK*17.0.12* : upgrading to subsequent patches is not discouraged in principle, as they generally contain fixes that do not disrupt compatibility. However, it should be noted that the Tagetik Data Loader is not beforehand tested with every JDK patch beyond version 17.0.12. This occurs only in response to specific requests or reports of issues identified after a JDK update.

E.g. linux: *export JAVA_HOME="/usr/lib/jvm/openjdk-17.0.12"*
E.g. windows *set JAVA_HOME=C:\Progra~1\Java\openjdk-17.0.12*

List of tested and supported JDKs:

- Amazon Corrected 17.0.12
- Microsoft 17.0.12
- Oracle 17.0.12
- SapMachine 17.0.12
- IBM Semeru 17.0.12
- Liberica 17.0.12
- Temurin 17.0.12

Definition of other Java and Karaf settings

To define the other Java and Karaf settings, it is necessary to access the *setenv* file *(setenv.bat* if installing on Windows operating systems) located in *<Karaf home>/bin* and add the options. If Karaf is run via Windows service, after running the wrapper installation command, configure the *wrapper.conf* file available inside the configuration directory. If a custom name is specified inside the wrapper installation command, the file will be *karaf-<name>.conf*

E.g. linux:

export JAVA_MIN_MEM=256M # Minimum memory for the JVM

export JAVA_MAX_MEM=1024M # Maximum memory for the JVM

export JAVA_PERM_MEM=128M # Minimum perm memory for the JVM

export JAVA_MAX_PERM_MEM=256M # Maximum memory for the JVM

E.g. windows

set JAVA_MIN_MEM=256M # Minimum memory for the JVM

set JAVA_MAX_MEM=1024M # Maximum memory for the JVM

set JAVA_PERM_MEM=128M # Minimum perm memory for the JVM

set JAVA_MAX_PERM_MEM=256M # Maximum memory for the JVM

Karaf configuration definition

It is recommended to externalise the Karaf configuration from the *<Karaf home>/etc* folder to another directory so that all customisations can be managed when updating the Karaf container. Once outsourced, it is possible to set up the configuration directory on Karaf by accessing the *setenv* file (*setenv.bat* if installing on Windows operating systems) located in *<Karaf home>/bin* and defining the KARAF_ETC.

E.g. linux: *export KARAF_ETC="/home/user/TagetikDataLoader/tgk-configuration"*

E.g. windows *set KARAF_ETC=C:\Progra~1\TagetikDataLoader\tgk-configuration*

General setting and connection to CCH Tagetik

To configure the TDL for both the connection to CCH Tagetik and to change certain parameters relating to its functionality, it is necessary to access the *tgk.properties.cfg* file in the Karaf configuration directory.

The parameters to be configured are:

- **tgk.agent.polling.interval**. Indicates at what intervals (in seconds) the TDL verifies requests for transactions to be taken over. It is recommended not to set up a value lower than 30 seconds
- **tgk.agent.tmpfolder**. Indicates the path reserved for the TDL to save temporary resources used for data manipulation. If the parameter is not specified, or the specified folder does not exist, then Tagetik Data Loader uses the Karaf's temporary folder, *<Karaf home>/data/tmp*, as the back-up folder. It is strongly recommended to specify this parameter, to avoid relying on temporary folders within the Karaf container.
- **tgk.agent.write.db.batchsize**. Indicates the number of statements sent simultaneously to the database to be executed at write time (default 10000)

- **tgk.agent.read.db.fetchsize**. Indicates the number of records to be retrieved simultaneously from the database during the reading phase (if not specified, the default value specified by the jdbc driver in use will be used).
  Note: the ideal value of this parameter may vary considerably depending on the database vendor and the complexity of the queries being executed.
- **tgk.endpoint.address**. CCH Tagetik endpoint URL (e.g. https://alpha.saastagetik.com/prod)

If secure data transfer is active, i.e. the Tagetik Endpoint URL has https as protocol, in the absence of specific parameterisation, the TDL will use the default JDK trustor in use. Should you wish to specify a custom server certificate, you can do so via the following parameters:

- **tgk.server.ks.password**. Password chosen when generating the server-side certificate. It must be written following the syntax specified in Security and encryption of sensitive data
- **tgk.server.ks.path**. Complete path of the server-side certificate

If the communication between TDL and CCH Tagetik is managed by a proxy server, the following can also be configured:

- **tgk.proxy.server**. Indicates the IP of the proxy server
- **tgk.proxy.port**. Indicates the proxy server port
- **tgk.proxy.user**. Indicates the user accessing the proxy server
- **tgk.proxy.password**. Indicates the password with which the user defined in the previous property can access the proxy server. It must be written following the syntax specified in Security and encryption of sensitive data


It is possible to enable payload encryption of data exchanged between the TDL and CCH Tagetik via PGP algorithm.

To do this, it is necessary to configure:

- **tgk.pgp.publickey** indicates the full path to the file containing the public key
- **tgk.pgp.privatekey** indicates the full path to the file containing the private key
- **tgk.pgp. passphrase** indicates the passphrase to use the private key. It must be written following the syntax specified in Security and encryption of sensitive data

If the Endpoint is only used to read data, only the *public key* needs to be configured; if it is only used to write data, only the secret *key* and *secret passphrase* need to be configured.
If the TDL is used to write data, only the *privatekey* with the *passphrase* on the TDL and the *publickey* on the Endpoint defined on CCH Tagetik need to be configured.

It is also possible to specify more technical properties to guide the operation of the Tagetik Data Loader, which can be particularly useful when analysing incorrect behaviour.

- **tgk.debug.polling.log.disabled**: This boolean property, when set to '*true*', disables logging of successful poll requests that did not trigger TDL actions. Failed poll requests, regardless of the error, and all other TDL actions, will continue to be logged. Enabling this property therefore makes the *tagetik.log* file much less verbose. The default value is '*false*'.
- **tgk.debug.tmp.cleanup.disabled**: This boolean property, when set to '*true*', disables the automatic deletion of temporary files generated by certain TDL operations. This option was introduced for diagnostic purposes, and allows you to analyse problems concerning the format or content of files generated by failed operations. The default value is '*false*'.

  
  Warning: This property must be used with caution, as enabling it may lead to the retention of numerous temporary files and this may cause considerable use of disk space.

Example 1 (HTTPS secure transfer without specified custom server certificates)

```
tgk.agent.polling.interval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
```

```
tgk.endpoint.address=https://tagetik.installation.url.com/tagetikcpm
tgk.server.ks.path=
tgk.server.ks.password=
```

Example 2 (HTTPS secure transfer with specified custom server certificates)

```
tgk.agent.polling.interval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=https://tagetik.installation.url.com/tagetikcpm
tgk.server.ks.path=/home/TDL/certificates/server.truststore
tgk.server.ks.password={TGK_ENCRYPT}server_side_certificate_password{TGK_ENCRYPT}
```

***Example 3* (HTTP transfer)**

```
tgk.agent.polling.interval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=http://tagetik.installation.url.com/tagetikcpm
```

***Example 4* (transfer with PGP)**

```
tgk.agent.polling.interval=30
tgk.agent.tmpfolder=/home/user/TagetikDataLoader/tgk-configuration/tmp/
tgk.agent.write.db.batchsize=10000
tgk.agent.read.db.fetchsize=
tgk.endpoint.address=http://tagetik.installation.url.com/tagetikcpm
tgk.pgp.publickey=/home/user/TagetikDataLoade/publickey.asc
tgk.pgp.privatekey=/home/user/TagetikDataLoade/privatekey.asc
tgk.pgp.passphrase={TGK_ENCRYPT}123_pippo{TGK_ENCRYPT}
```

Authentication configuration

To enable the communication between TDL and CCH Tagetik, it is necessary to define the authentication parameters. They can be found in the file *tgk.agent.cfg* in the Karaf configuration directory.

The parameters are as follows:

- **tgk.agent.id**. Unique ID associated with the TDL instance. It is automatically generated upon the first launch of Karaf.
- **tgk.agent.secret**. SecretKey to authenticate the TDL on CCH Tagetik. It must be defined after the first launch of Karaf, once the ID has been automatically generated, and must be defined using the syntax specified in Security and encryption of sensitive data

With these two parameters it is possible to register the TDL on CCH Tagetik and authenticate it upon each request.

Log configuration

The TDL log system is nothing more than a customization of the log system of Apache Karaf. The log configuration can be found in the file *org.ops4j.pax.logging.cfg* in the Karaf configuration directory and is a standard Log4j configuration file.

This file contains the following section with the specified values:

```
#Tagetik agent appender
log4j2.logger.tagetik.name = com.tagetik.cpm
log4j2.logger.tagetik.level=INFO
log4j2.logger.tagetik.additivity=false
log4j2.logger.tagetik.appenderRef.TagetikRollingFile.ref = TagetikRollingFile
log4j2.appender.tagetik.type = RollingRandomAccessFile
log4j2.appender.tagetik.name = TagetikRollingFile
log4j2.appender.tagetik.fileName = ${karaf.data}/log/tagetik.log
log4j2.appender.tagetik.filePattern = ${karaf.data}/log/tagetik.log.%i
log4j2.appender.tagetik.append = true
log4j2.appender.tagetik.layout.type = PatternLayout
log4j2.appender.tagetik.layout.pattern = ${log4j2.pattern}
log4j2.appender.tagetik.policies.type = Policies
log4j2.appender.tagetik.policies.size.type = SizeBasedTriggeringPolicy
log4j2.appender.tagetik.policies.size.size = 16MB
```

For the meaning of the various options and of the values to be assigned, refer to the Log4j specifications.

Log files are saved by default in the *<Karafhome>/data/log/* folder

Based on the default settings above, the TDL log file is named *tagetik.log* and is generated on a daily basis. Moreover, every day the file generated on the previous day is renamed by adding a suffix given by the production date.

Datasource configuration

In case database operations are required such as:

- reading data to be transferred to CCH Tagetik
- writing of data extracted from CCH Tagetik

the TDL requires the creation and configuration of *tgk.datasource-<custom_name>.cfg* files in the Karaf configuration directory.

The Karaf configuration folder (tgk-configuration) also contains the vendor templates officially supported by the Tagetik Data Loader.

For authentication with *basic authentication*:

- **SqlServer (JTDS driver)**: *_tgk.datasource-jtds_template.cfg*
- **SqlServer (Microsoft driver)**: *_tgk.datasource-microsoft_template.cfg*
- **Oracle**: *_tgk.datasource-oracle_template.cfg*
- **PostgreSQL**: *_tgk.datasource-postgresql_template.cfg*
- **SAP Hana**: *_tgk.datasource-hana_template.cfg*
- **IBM DB2**: *_tgk.datasource-db2_template.cfg*
- **Netsuite**: *_tgk.datasource-netsuite_template.cfg*
- **Snowflake**: *_tgk.datasource-snowflake-basicAuth_template.cfg*

For the *Snowflake* vendor, authentication is also managed via:

- **Key Pair**: *_tgk.datasource-snowflake-keyPair_template.cfg*
- **OAuth2.0**: *_tgk.datasource-snowflake-oAuth_template.cfg*


Warning: With regard to authentication with *Key Pair*, please refer to the Snowflake documentation for certificate generation and management.
As far as *OAuth2.0* is concerned, only the grant type '*client credentials*' is accepted.
For configuration and further details on the *Snowflake* side for both types of authentication, please refer to the official Snowflake documentation

 The password type fields of the datasources, ('password' for basic authentication, 'private_key_pwd' and 'oauthClientSecret' for Snowflake authentication) must be entered (both when defining and editing) with the syntax specified in Security and encryption of sensitive data

Adding a custom datasource

From the Karaf command line via the command *service:list DataSourceFactory* it is possible to display the installed JDBC drivers.

To add a new JDBC driver it is necessary to:

- copy the **[driver_name].jar** into the *<Karaf home>/lib/endorsed* folder
- run the *install -s wrap:file:./lib/endorsed/[driver_name].jar* command from the Karaf command line
- add a *.cfg* file relating to the new datasource into the Karaf configuration directory (like those described above)
- restart Karaf

 The simultaneous presence of two JDBC drivers from the same vendor causes problems at runtime. If it is necessary to install a new JDBC driver (E.g. : a later version), the previous one has to be uninstalled.

Path configuration

In case operations are needed on files such as:

- file transfer to CCH Tagetik
- writing files produced by CCH Tagetik

TDL requires the *tgk.repository.cfg* file to be created and configured in the Karaf configuration directory.

Inside, a mapping list between the logical path name and the real absolute path can be configured.

The syntax is typical of the *.properties files, thus:

<path-1-logical-name>=<path-1-value>

<path-2-logical-name>=<path-2-value>

Security and encryption of sensitive data

For keystore and datasource passwords, there is a mechanism for encrypting the values.

To define the value of the password, the following syntax as to be used: *<password_property_name>={TGK_ENCRYPT}<password_value_to_encrypt>{TGK_ENCRYPT}*

After saving the configuration file, Tagetik Data Loader will automatically encrypt the value of the property and replace it in the same configuration file.

 TDL can only handle passwords containing 'ASCII Printable Characters', i.e. ASCII characters 32 to 126 with the following exceptions:

- The '$' character (ASCII 36) cannot be used as the last character of the password
- The character '\' (ASCII 92) cannot be used in the password.

Starting Karaf

Once installation and configuration have been carried out, Tagetik Data Loader can be used by starting Karaf via the *karaf* command(*karaf.bat* if installation is carried out on Windows operating systems) present in *<Karaf home>/bin*.

Starting as an operational system service

It is also possible to start Karaf as an OS service by executing the command *wrapper:install --name "TDL" --display "TDL" --description "TDL"* from the Karaf command line and afterwards:

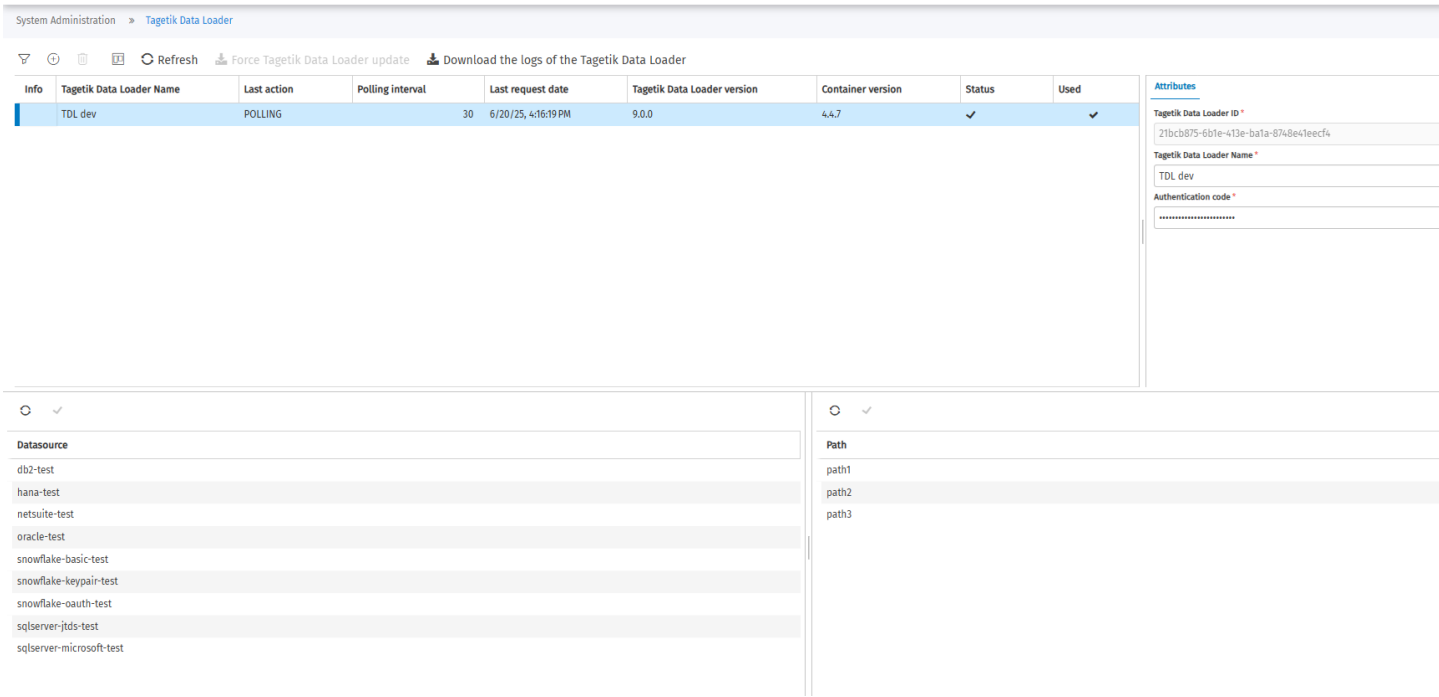
- for Windows: run from the command prompt the command *TDL-service.bat install* (or *karaf-service.bat install*), present in the *<Karaf home>/bin* directory
- for Linux: follow the instructions shown once you run the *wrapper:install* command .

Tagetik Data Loader - Registration and monitoring on CCH Tagetik

CCH Tagetik provides the admin users with a window to:

- register new TDLs so that CCH Tagetik enables communication with them
- remove TDLs no longer active and not used in any parametrization (Endpoint)
- display the information and the status of the configured TDLs in real time
- display the last or current TDL operation performed
- request the manual update of a TDL if the automatic update fails
- request to download TDL logs
- request the updating of the list of datasources configured on the TDL and their verification
- request the updating of the list of paths configured on the TDL and their verification

This monitor functionality is accessible from the tile of the same name available on the Repository home page.



The configured TDLs are displayed at the top of the window, and for each one is shown:

- **Tagetik Data Loader Name**: TDL description defined by the user
- **Last action**: last command executed by the TDL or in progress
- **Polling interval**: specifies at what intervals (in seconds) the TDL checks for operations to be taken over
- **Last requested** date: date of the last 'heart beat' made by the TDL
- **Tagetik Data Loader version**: TDL version installed on Karaf

- **Container** version: Karaf container version
- **Status**: displays the status of the TDL via an indicator:
  - ✓ : on-line
  - ✗ : off-line
  - ⚠ : on-line but locked
    - Authentication failed
    - Waiting for automatic update
    - Automatic update in progress
    - Automatic update failed
    - Manual upgrade required
- **Used**: indicates whether the TDL is used in one of the configured Endpoints.

There are two different sections at the bottom of the window:

- **Datasource**: where the system displays the Datasources' logical names,
- **Path:** where the system displays the logical names of Paths

both configured on Karaf for the current TDL. For both sections there are the following buttons:

- ↻ : allows updating data in the section by retrieving the information from the configuration files of the current TDL. This button is only enabled if the selected TDL is online
- ✓ : verifies the validity of the selected Datasource/Path. The result of the verification, whether positive or negative, is notified at the end of the operation by an alert containing an appropriate message.

Registration of a new TDL

CCH Tagetik only enables communication with a TDL if it has been previously registered.

A new TDL can be registered via the button ⊕. The system enables the *Attributes* panel where you must specify:

- **Tagetik Data Loader ID**: unique identifier of the TDL. The field must be filled with the ID automatically generated by Karaf when the TDL is first started in the *tgk.agent.cfg* file in the Karaf configuration directory.
- **Tagetik Data Loader Name**: TDL description defined by the user
- **Authentication** code: identifies the authentication code associated with the TDL; it must match the one specified in the *tgk.agent.cfg* file in the Karaf configuration directory.

TDL management

Once a TDL has been registered, you can:

- edit the **Tagetik Data Loader Name** and **Authentication Code** via management in detail,
- refresh the displayed information (e.g. **Status**) via the button ↻ Refresh
- remove the configuration of a TDL via the 🗑 button.

Editing and deletion are only possible if the TDL is not used in a parametrization of an Endpoint.

TDL update

The version of a TDL (e.g. 6.0.1) consists of three pieces of information:

- *Major*: used to identify 'breaking-change' such as changing the Karaf container or changing the communication protocol,
- *Minor*: used to identify the addition of new features that do not compromise the use of existing ones,
- *Patch*: used to identify corrections to existing functionalities.

How it is updated depends on the difference between the version of the reference TDL on CCH Tagetik and the version of the TDL on Karaf :

- If the two *majors* are different, the **Status of** the Tagetik Data Loader will be changed to (warning) and *TO_UPGRADE* will be displayed as **Last Action**. This means that a manual update of the Karaf/TDL container will have to be carried out.
- if the *Minor.Path* of the TDL on CCH Tagetik is smaller than that of the TDL on Karaf, no update is necessary.
- if the *Minor.Path* of the TDL on CCH Tagetik is greater than that of the TDL on Karaf, an automatic update will be performed. In the **Last Action** field the commands *TO_UPDATE*, *UPDATING* and, if the update was successful, the POLLING command (which certifies that communication is active again) will be performed, otherwise the *UPDATE_FAILED* command will be displayed. In this case, a manual update can be retried via the Force Tagetik Data Loader update button.

Download Tagetik Data Loader Logs

Starting with version 9.0.0 of Tagetik Data Loader, it is possible to download the logs generated by the TDL directly from CCH Tagetik, from the Agent configuration window via the

Download the logs of the Tagetik Data Loader button, generating a zip file containing all the files in the log folder configured on Karaf. This feature is only enabled for TDL with version greater than or equal to 9.0.0 and On-line status.

Configuration on CCH Tagetik

In the configuration of the CCH Tagetik Repository, there are settings relating to the use of TDL:

Data write: data packet size (Kbytes): Indicates the maximum size in KBytes into which data extracted from CCH Tagetik to be transmitted to Tagetik Data Loader must be divided and sent.

Read Data : Data Packet Size (Kbytes): Indicates the maximum size in KBytes into which data extracted from Tagetik Data Loader is to be divided and sent to CCH Tagetik.

Time out (min): Indicates the minutes CCH Tagetik waits for the Tagetik Data Loader to respond to a command. After this time has passed, a 'Timeout' error is returned by the application. The default time is 30 minutes.

Important: As of Tagetik Data Loader version 9.0.0, this time out will also be applied on the TDL on premise, also interrupting the current thread.

Endpoint

to conclude the configuration of the TDL, it is necessary to specify an agent-type endpoint to be used with ETL and Quick Data Loader (see Endpoint)